Periodic Quotient Graphs, Multifractal Spectra and a Dimensional Theory of Morphism for Metal–Organic Frameworks: Exact Constructions, Verified Invariants and a Finite-Size Obstruction

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Abstract

Metal–organic frameworks (MOFs) are modular crystals whose function is governed by how metal clusters and organic linkers are wired into a periodic network. We ask whether a descriptor computed from *connectivity alone*, with no molecular simulation, can summarise that wiring, and we treat the question as a mathematical one. A MOF is formalised as a voltage graph (Γ, ν) with voltages in $\mathbb{Z}^3$; we prove that coarse-graining by the metal-oxo rule commutes with supercell lifting, and verify the construction on HKUST-1, MOF-5, ZIF-8 and UiO-66 against published coordination numbers $(4, 6, 4, 12)$. We derive the Bloch decomposition of the supercell Laplacian, show that its algebraic connectivity decays as n^{-2} and that $n^2\lambda_2$ converges to $4\pi^2$ times a cell-independent homogenised stiffness, confirmed numerically to 0.1–1%. For the multifractal descriptor of Xiao *et al.* we prove that the through-origin estimator gives a concave $\tau(q)$ and hence an inverted-parabola $f(\alpha)$ with apex $D_0 = \rho_D \ln|I|$, whereas a free-intercept fit pins $\tau(0) = 0$, a defect we found and corrected. An exact tail identity $\alpha(-\infty) = \rho_D \ln N - \mu_D$ then explains, to better than 0.4%, why the spectrum width $\Delta\alpha$ grows without bound with supercell size, and why two bands we had computed must be withdrawn. Finally we propose a dimensional theory of framework *morphism*: a periodicity rank that steps $3 \to 2 \to 1 \to 0$ under linker deletion (verified on MOF-5), together with shape invariants that classify building-block cores as tetrahedral (Zn_4), octahedral (Zr_6) or rod-like (Cu_2). We separate clearly what is proved, what is verified numerically, and what is not claimed.

Keywords: metal–organic frameworks; periodic graphs; voltage graphs; graph Laplacian; Bloch decomposition; multifractal analysis; persistent homology; framework morphism.
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1 Introduction

Metal–organic frameworks (MOFs) are crystalline solids assembled from metal-containing clusters (*nodes*) joined by organic struts (*linkers*) [8, 11]. Changing either part changes the pore network and with it the properties that matter for gas storage, separation and catalysis. The design space is enormous: hundreds of thousands of hypothetical and experimental structures are catalogued [5, 24], yet deciding whether a candidate deserves a molecular simulation (hours per structure) is the bottleneck of computational screening.

Recent work shows that the *representation* of a framework decides how much a model can learn. Graph networks encode atoms and bonds [26]; the Molecular Graph Transformer of Anselmi *et al.* adds a line graph for bond angles and a Coulomb-matrix graph for long-range electrostatics [1]; the interaction topological transformer (ITT) of Chen *et al.* feeds persistent-homology features of element clusters into a transformer and reaches state of the art on 17 of 18 adsorption, transport and stability benchmarks [4]. These are learned, data-hungry models. Here we ask the complementary, purely mathematical question:

> *Can a descriptor computed from a framework's connectivity alone, in seconds and with no molecular simulation, carry enough information to rank or group frameworks, and what can be proved about it?*

Approach. We model a MOF as a finite graph Γ whose edges carry integer translations $\nu(e) \in \mathbb{Z}^3$ (a *voltage graph* in the sense of Gross and Tucker [12]). The infinite crystal is the derived graph; any finite supercell is a covering of Γ (Section 3). On this object we study (i) the graph Laplacian and its Bloch decomposition, (ii) the multifractal spectrum of Xiao, Wang and Li [25], and (iii) a periodicity rank and shape invariants that formalise what it means for a framework to *morph* between one-, two- and three-dimensional (and discrete) forms.

Contributions.

C1. Exact construction. Coarse-graining by the metal-oxo rule [3] commutes with supercell lifting (Theorem 4.1); a pipeline built on it reproduces the published coordination numbers of four benchmark nets and a full-graph-isomorphism check on five lifted cases.

C2. Spectral theory of the supercell. A closed-form Bloch decomposition of the supercell Laplacian (Theorem 5.1); the algebraic connectivity obeys $\lambda_2(\mathcal{L}_n) \sim n^{-2}$, and $n^2\lambda_2$ converges to $4\pi^2$ times a cell-independent stiffness tensor Σ (Theorem 5.3 and Proposition 5.5), confirmed to 0.1–1%.

C3. Structure of the multifractal estimator. With the through-origin definition of $\tau(q)$, the spectrum is necessarily an inverted parabola with apex $D_0 = \rho_D \ln|I|$; the common free-intercept shortcut pins $\tau(0) = 0$ (Proposition 6.2). We found and fixed this defect in our own code.

C4. An exact finite-size obstruction. The left tail of the spectrum obeys $\alpha(-\infty) = \rho_D \ln N - \mu_D$ (Theorem 6.3); since μ_D stays bounded, the spectrum width diverges with system size and cannot be a material property. This explains quantitatively why we withdraw two earlier "bands".

C5. A dimensional theory of morphism. A periodicity rank ϱ that is monotone under linker deletion and steps $3 \to 2 \to 1 \to 0$ on MOF-5 (Theorem 8.3); gyration- and third-moment invariants that classify the building-block cores as tetrahedral, octahedral and rod-like in agreement with crystallography (Section 8.2).

What this paper does not claim. The descriptors here are blind to element identity and to geometry beyond lattice vectors. Nothing is computed about adsorption, selectivity or stability, and none of the statements about morphism is a statement about crystal growth kinetics. We mark every result as **[proved]**, **[verified]** (checked numerically on real structures, with the script named in Appendix A) or **[proposed]**. Negative results are reported as such (Section 7).

Organisation. Section 2 positions the work against [1, 4]. Sections 3 and 4 set up voltage graphs and the CIF-to-graph construction. Sections 5 and 6 develop the spectral and multifractal theory. Section 7 collects the numerical evidence including the withdrawn claims, Section 8 develops the morphism theory, and Section 9 discusses limitations and the route to chemical properties.

2 Related work and positioning

Persistent interaction topology and the ITT [4]. Chen, Liu, Chen and Wei generalise the simplicial complex to an *interaction complex* $\mathcal{K} = (K, \{K_i\}_{i=1}^n)$, where K is a simplicial complex on all atoms and $K_i \subseteq K$ are subcomplexes induced by chemically chosen element clusters. An *interaction simplex* is a tuple $(\sigma_1, \ldots, \sigma_n)$ with $\sigma_i \in K_i$ and $\sigma_1 \cap \cdots \cap \sigma_n \neq \emptyset$. Interaction homology $H_p(\{K_i\}; \mathbb{K})$ and its persistent Betti numbers $\beta_p^{a,b} = \dim_{\mathbb{K}} H_p^{a,b}[\mathcal{F}; \mathbb{K}]$ are computed along an α-complex filtration. For $n = 1$ this reduces to ordinary persistent homology [9, 27]. The ITT uses four token types—1 structural, 7 elemental (cluster C_0–C_6), up to 256 atomic, and $7 \times 6 = 42$ pairwise interaction tokens—each topological token being a vector of Betti numbers sampled on a grid from 0 to 25 Å in steps of 0.1 Å (so $250 \times 3 = 750$ numbers for $p = 0, 1, 2$). After self-supervised pretraining on ≈ 0.6 million porous structures, it gives state-of-the-art predictions on 17 of 18 tasks, and saliency analysis shows that adsorption and diffusion rely more on the topological tokens while band gap relies more on atomic tokens.

Relation to our work. Our voltage graph Γ is the combinatorial 1-skeleton at the building-block scale; its Betti numbers β_0, β_1 (Table 3) are the discrete analogues of H_0, H_1 at the linker scale. The differences are instructive. (i) They work in $\mathbb{R}^3$ on a supercell normalised to about 64 Å (as stated in their preprocessing), whereas we treat periodicity *exactly* through voltages and obtain closed-form spectral statements (Section 5). (ii) They are element-aware through the clusters C_0–C_6; our graph is deliberately untyped. (iii) They learn from data; we define invariants and prove properties of them. The two views are complementary, and Section 9 outlines how interaction homology of the *periodic* complex could be computed on Γ.

DFT-validated ML screening [1]. Anselmi, Slabaugh, Crespo-Otero and Di Tommaso screen 20 375 QMOF structures [19] for CO_2-reduction photocatalysts. PBE calculations select 2169 candidates, hybrid-DFT (HSE06) yields 105 hits (this step took 24 CPU-days), and the MGT model reproduces band gaps and band edges in 4.5 hours with a mean error of 0.34 eV. Three findings are relevant here. First, prediction errors are organised by chemistry, not random: multi-metal centres, heavy d/f metals and halides are worst, and lanthanides show 0.18 eV error on seen metals but 1.09 eV on unseen ones, i.e. memorisation. Second, widening the acceptance window by the model error ($\delta = 0.34$ eV) raises the recovery of DFT hits from 20% to 69% (98% at $\delta = 1.0$ eV, at the price of selecting 72% of the set). Third, the frontier-orbital positions are not determined by the linker alone; the metal matters. *Relevance:* the error-margin device is exactly how a cheap descriptor should be used as a pre-filter (Section 9), and finding three is a reminder that a connectivity-only descriptor is, by construction, blind to an important factor. Their line graph (bond angles) carries geometric information that our graph discards.

Multifractal network analysis. Box-covering fractal dimensions of networks were introduced by Song, Havlin and Makse [20]; the multifractal formalism for measures goes back to Halsey *et*

Table 1. Representation levels used by recent MOF models and by this work.

Model	Mathematical object	Periodicity	Element-aware	Learned
CGCNN [26]	atom–bond graph	supercell/cutoff	yes	yes
MGT [1]	bond, line, Coulomb graphs	fully connected global graph	yes	yes
ITT [4]	interaction complexes, persistent Betti vectors	normalised supercell (≈ 64 Å)	yes (7 clusters)	yes
This work	voltage graph (Γ, ν) with $\nu \in \mathbb{Z}^3$	exact (derived graph, Bloch)	no (by design)	no (closed form)

al. [13]; Xiao, Wang and Li define (influential-)node multifractal analysis, iNMFA [25], which we implement. We use it only as a *candidate* descriptor and test it critically.

Decomposition, topology and crystal nets. Metal-oxo decomposition is the default algorithm of MOFid [3]; net identification is done with Systre and RCSR [7, 18]. The mathematics of periodic nets as abelian covers of finite graphs is the subject of topological crystallography [16, 22].

3 Mathematical preliminaries: voltage graphs and periodic nets

Throughout, a *graph* is a finite multigraph $\Gamma = (V, E, o, h)$ with vertex set V, edge set E, and tail/head maps $o, h\colon E \to V$; loops ($o(e) = h(e)$) and parallel edges are allowed. An *orientation* is chosen once for every edge; the reverse of e is written $\bar{e}$. The degree $\deg(v)$ counts edge-ends at v (a loop counts twice).

3.1 Voltage graphs and derived graphs

Definition 3.1 (Voltage graph and derived graph [12])**.** A *voltage assignment* is a map $\nu\colon E \to \mathbb{Z}^3$, extended by $\nu(\bar{e}) = -\nu(e)$. The pair (Γ, ν) is a *voltage graph*. Its *derived graph* Γ^ν has vertex set $V \times \mathbb{Z}^3$ and, for every $e \in E$ and $c \in \mathbb{Z}^3$, one edge (e, c) from $(o(e), c)$ to $(h(e), c + \nu(e))$.

The group $\mathbb{Z}^3$ acts on Γ^ν by $c' \cdot (v, c) = (v, c + c')$, freely on vertices and edges, and the quotient is Γ. The projection $p\colon \Gamma^\nu \to \Gamma$ is a regular covering with deck group $\mathbb{Z}^3$. This is the graph-theoretic form of a crystal: the *unit cell* is the finite base graph, the *lattice* is $\mathbb{Z}^3$, and the voltage $\nu(e)$ records which neighbouring copy of the cell the edge e reaches (cf. Figure 1).

Definition 3.2 (Periodic net)**.** A *periodic net* is the derived graph $X = \Gamma^\nu$ of a voltage graph with Γ finite. For $n \geq 1$ the *n-supercell* $G_n := \Gamma^{\nu \bmod n}$ is the derived graph with voltages reduced in $\mathbb{Z}^3_n = (\mathbb{Z}/n\mathbb{Z})^3$: it has vertex set $V \times \mathbb{Z}^3_n$, $|V(G_n)| = n^3\,|V|$ and $|E(G_n)| = n^3\,|E|$. It is the torus quotient of X by the sublattice $n\mathbb{Z}^3$.

Proposition 3.3 (Covering facts [proved])**.** *Let $p_n\colon G_n \to \Gamma$ be the projection. Then (a) $\deg_{G_n}(v, c) = \deg_\Gamma(v)$ for all (v, c); (b) if Γ is bipartite with parts V_N, V_L then so is G_n; (c) $\mathbb{Z}^3_n$ acts on G_n by automorphisms, freely, with quotient Γ; (d) G_n has $|V|\,n^3$ vertices and every vertex of G_n lies over exactly one vertex of Γ.*

Proof. For (a), the edges at (v, c) are (e, c) for e with $o(e) = v$ and $(e, c - \nu(e))$ for $h(e) = v$; these are in bijection with the edge-ends of Γ at v. (b) and (c) follow because the second coordinate is ignored by the colouring and translated freely by the action. (d) is the definition. □

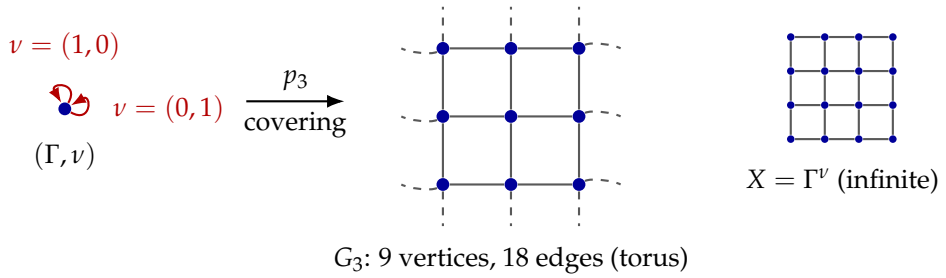


Figure 1. A voltage graph and its covers. The base graph (Γ, ν) has one vertex and two loops carrying voltages $(1,0)$ and $(0,1)$ (a two-dimensional illustration of the three-dimensional case). Its derived graph X is the square lattice; reducing voltages modulo 3 gives the 3×3 torus supercell G_3, a 9-fold regular covering. The numerical pipeline of Section 4 builds (Γ, ν) from a CIF file and then forms G_n combinatorially.

Definition 3.4 (Net voltage group and periodicity rank)**.** Let Γ be connected. For a closed walk $W = e_1 \cdots e_m$ put $\nu(W) = \sum_i \nu(e_i) \in \mathbb{Z}^3$. The *net voltage group* $H(\Gamma, \nu) = \{\nu(W)\} \leq \mathbb{Z}^3$ is a subgroup (closed walks concatenate) and, since $\mathbb{Z}^3$ is abelian, does not depend on the base vertex. The *periodicity rank* is $\varrho(\Gamma, \nu) := \operatorname{rank}_{\mathbb{Z}} H(\Gamma, \nu) \in \{0, 1, 2, 3\}$. For disconnected Γ, ϱ is the vector of ranks of its components.

Proposition 3.5 (Components of the derived graph [proved])**.** *For connected Γ, the derived graph $X = \Gamma^\nu$ has exactly $[\mathbb{Z}^3 : H]$ connected components if* rank $H = 3$*, and infinitely many otherwise; each component is invariant under a free action of $H \cong \mathbb{Z}^\varrho$ with finite quotient Γ. In particular $\varrho = 3$ if and only if X is connected up to finite index, and $\varrho = 0$ if and only if every component of X is isomorphic to Γ itself (a* discrete *fragment).*

Proof. Fix a spanning tree T and define $\pi(v) \in \mathbb{Z}^3$ as the voltage along the tree path from a base vertex to v. A vertex (v, c) lies in the component of $(v_0, 0)$ iff $c - \pi(v) \in H$: walking a closed loop at v_0 shifts the second coordinate by the loop's net voltage, and every element of H is realised; conversely, any walk from $(v_0, 0)$ to (v, c) projects to a walk from v_0 to v, which differs from the tree path by a closed walk. Hence the components are indexed by the cosets $\mathbb{Z}^3 / H$, each acted on freely by H with quotient Γ. □

3.2 The Laplacian

For a finite graph G with oriented edges, the *incidence vector* of e is $b_e = \mathbf{e}_{o(e)} - \mathbf{e}_{h(e)} \in \mathbb{R}^V$ and the *graph Laplacian* is

$$\mathcal{L}(G) = \sum_{e \in E} b_e b_e^{\mathsf{T}} = D - A, \qquad \langle \psi, \mathcal{L}\psi \rangle = \sum_e |\psi(o(e)) - \psi(h(e))|^2 . \tag{1}$$

(Loops contribute $b_e = 0$.) It is symmetric positive semidefinite, with eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_{|V|}$. The multiplicity of 0 equals the number of connected components, and λ_2, the *algebraic connectivity*, is positive iff G is connected [10, 17].

Proposition 3.6 (Interlacing under edge and vertex deletion [proved, verified])**.** *Let G' be obtained from G by deleting a set S of s edges. Then for all k*

$$\lambda_{k-s}(\mathcal{L}(G)) \;\leq\; \lambda_k(\mathcal{L}(G')) \;\leq\; \lambda_k(\mathcal{L}(G)), \qquad \lambda_k(\mathcal{L}(G)) - \Big\|\textstyle\sum_{e \in S} b_e b_e^{\mathsf{T}}\Big\| \leq \lambda_k(\mathcal{L}(G')). \tag{2}$$

If $G - v$ is obtained by deleting a vertex v of degree d, then spec $\mathcal{L}(G - v) \cup \{0\} =$ spec $\mathcal{L}(G_v)$ *where G_v is G with the d edges at v removed, and $\big\|\sum_{e \ni v} b_e b_e^{\mathsf{T}}\big\| = d + 1$ for a simple graph, so $\lambda_k(G) - (d+1) \leq \lambda_k(G_v) \leq \lambda_k(G)$.*

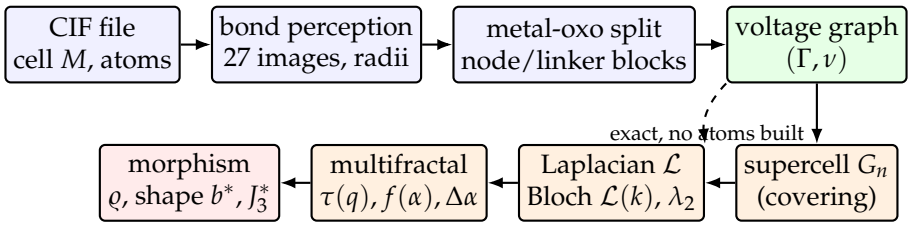


Figure 2. Pipeline. Stages 1–3 are a translation of MOFid's metal-oxo decomposition [3]; the voltage graph is the project's own construction. Everything to the right of (Γ, ν) is computed *combinatorially*, without rebuilding atoms.

Proof. $\mathcal{L}(G') = \mathcal{L}(G) - P$ with $P = \sum_{e \in S} b_e b_e^{\mathsf{T}} \succeq 0$ of rank $\leq s$. Weyl's inequalities [14, §4.3] give $\lambda_k(\mathcal{L} - P) \leq \lambda_k(\mathcal{L})$ and $\lambda_k(\mathcal{L} - P) \geq \lambda_{k-s}(\mathcal{L})$ for a positive semidefinite P of rank s, and $\lambda_k(\mathcal{L} - P) \geq \lambda_k(\mathcal{L}) - \lambda_{\max}(P)$. The vertex statement: G_v is $G - v$ plus an isolated vertex, whose Laplacian block is 0; the star $\sum_{e \ni v} b_e b_e^{\mathsf{T}}$ is the Laplacian of $K_{1,d}$ whose largest eigenvalue is $d + 1$. □

We confirmed Equation (2) numerically by deleting 200 random edges one after the other from the $2 \times 2 \times 2$ MOF-5 supercell: zero violations (`check_03_morphism_ladder.py`).

4 From a CIF file to a voltage graph

4.1 Pipeline overview

4.2 Bond perception

Let the cell matrix $M \in \mathbb{R}^{3\times3}$ have the lattice vectors as rows, so that lattice coordinates $t \in \mathbb{Z}^3$ correspond to Cartesian displacement $t^{\mathsf{T}}M$; write $\Phi(t) = t^{\mathsf{T}}M$. For atoms a, b with Cartesian positions x_a, x_b, the *minimum-image displacement* is

$$v_{ab} = x_b - x_a + \Phi(s^*), \qquad s^* = \operatorname*{arg\,min}_{s \in \{-1,0,1\}^3} \|x_b - x_a + \Phi(s)\|, \tag{3}$$

found by searching the 27 neighbouring images. The pair is declared bonded if $0.4 < \|v_{ab}\| < \kappa_{ab}(r_a + r_b)$, with r the covalent radii of Cordero *et al.* [6] and $\kappa_{ab} = 1.30$ if a or b is a metal and 1.15 otherwise (the larger factor accommodates longer metal–ligand bonds). The result is the *bond voltage graph* (B, ν_B) of the unit cell, with atoms as vertices.

4.3 Metal-oxo decomposition

Let $A_{\mathrm{M}} \subseteq A$ be the metal atoms (determined by exclusion: every element not in a list of 24 non-metals). Let B^- be B with every bond incident to A_{M} deleted and let $\mathcal{C}$ be its connected components ("fragments"). Each fragment is labelled

$$\mathrm{role}(C) = \begin{cases} \text{node} & \text{if } |C| = 1 \text{ or } C \subseteq \{\mathrm{O}, \mathrm{H}\}, \\ \text{linker} & \text{otherwise,} \end{cases} \tag{4}$$

(a lone atom is a metal stripped of all ligands; an O/H-only fragment is an oxo/hydroxo bridge, e.g. the μ_4-O at the centre of Zn_4O). *Node blocks* are the connected components, in the *original* bond graph B, of the set of node-role atoms; *linker blocks* are the components, in B, of the set of linker-role atoms. The rule cuts every carboxylate from the metal, so the reported node is the bare metal core (HKUST-1 $\to Cu_2$, MOF-5 $\to Zn_4O$, UiO-66 $\to Zr_6O_8H_4$ in this implementation). This changes block *composition* but not which blocks are joined, which is all the graph records.

Algorithm 1 Voltage graph of a MOF from its CIF file.

Require: cell matrix M, atoms (s_a, x_a)
$(B, v) \leftarrow$ BONDPERCEPTION(M, x, s) ▷ 27-image minimum-image search
$B^- \leftarrow B$ minus all bonds incident to metals; $\mathcal{C} \leftarrow$ components(B^-)
label every $C \in \mathcal{C}$ as node or linker (singleton or $\{\mathrm{O}, \mathrm{H}\}$-only $\Rightarrow$ node)
blocks $\leftarrow$ components of B restricted to node atoms $\cup$ components of B restricted to linker atoms
for each block β **do**
    unwrap β by BFS to obtain u_a for $a \in \beta$
end for
$E \leftarrow \emptyset$
for each bond (a, b) with $\beta(a) \neq \beta(b)$ or non-zero voltage **do**
    $t \leftarrow \mathrm{round}\big((u_a + v_{ab} - u_b)M^{-1}\big)$; $E \leftarrow E \cup \{\mathrm{canon}(\beta(a), \beta(b), t)\}$
end for
return $(V = \text{blocks}, E)$ with $\nu(i, j, t) = t$

4.4 Extracting the voltages

A block that straddles a cell face has atoms whose Cartesian coordinates differ by a lattice vector from their true neighbours. We therefore *unwrap* every block: a BFS over its internal bonds assigns each atom a a position u_a consistent with one choice of images, using the stored displacements v. Because blocks are finite, this is consistent (no cycle of a block has non-zero net voltage; this is the hypothesis used in Theorem 4.1). For a bond (a, b) between different blocks $i \ni a$, $j \ni b$, the two frames differ by an integer lattice vector

$$t_{ab} = \mathrm{round}\big((u_a + v_{ab} - u_b)\, M^{-1}\big) \in \mathbb{Z}^3, \tag{5}$$

which is the voltage of the edge $i \to j$. Edges are identified by the triple (i, j, t) with canonical orientation $(i, j, t) \sim (j, i, -t)$; distinct atomic bonds giving the same triple (e.g. the two oxygens of one carboxylate binding one cluster) collapse to a single edge. A self-loop (i, i, t) with $t \neq 0$ is kept (a block bonded to its own periodic image); a bond $(i, i, 0)$ is internal and dropped. The output is (Γ, ν) with $V = \{\text{blocks}\}$, $E = \{(i, j, t)\}$. Algorithm 1 summarises the procedure.

4.5 Coarse-graining commutes with supercell lifting

The pipeline builds G_n from (Γ, ν) without atoms. One may ask whether this agrees with the "obvious" alternative—build the atomic n-supercell first, then coarse-grain. The answer is yes, and the proof is short.

Theorem 4.1 (Coarse-graining commutes with lifting [proved, verified])**.** *Let (B, ν_B) be a bond voltage graph whose metal-oxo fragments and node/linker blocks are all finite, i.e. every block has trivial net voltage group. Let K denote metal-oxo coarse-graining. Then for every $n \geq 1$,*

$$K\big(B^{\nu_B \bmod n}\big) \;\cong\; K(B)^{\nu \bmod n} = G_n,$$

i.e. coarse-graining the atomic supercell gives exactly the n-fold cover of the coarse-grained cell.

Proof. Write $\tilde{B} = B^{\nu_B \bmod n}$ and let $p\colon \tilde{B} \to B$ be the covering projection. (1) Deleting the bonds incident to metals commutes with p: the set of deleted edges in $\tilde{B}$ is exactly p^{-1} of the deleted set in B (being incident to a metal depends only on the element labels of the endpoints, which p preserves). (2) Let C be a component of B^-. It has trivial net voltage by hypothesis, so by Proposition 3.5 (applied with $H = 0$) its preimage in $\tilde{B}^-$ consists of exactly n^3 pairwise disjoint copies $C \times \{c\}$, each isomorphic to C, and every component of $\tilde{B}^-$ arises this way. (3) The role of

Table 2. Empirical check of Theorem 4.1: decompose-then-replicate (Route A) versus replicate-then-decompose (Route B, with bond perception redone on the atomic supercell). Isomorphism was tested with `networkx.is_isomorphic`. Timing is for one run on the authors' workstation; only the ratio is meaningful.

Structure	n	vertices / edges of G_n	degree sequences	isomorphic	cost ratio B/A
HKUST-1	2	112 / 192	equal	yes	57.0×
MOF-5	2	256 / 384	equal	yes	59.9×
ZIF-8	2	288 / 384	equal	yes	64.7×
UiO-66	2	56 / 96	equal	yes	59.1×
UiO-66	3	189 / 324	equal	yes	590.2×

Table 3. Benchmark voltage graphs. $\beta_1 = |E| - |V| + 1$ is the cyclomatic number of the (connected) base graph; ϱ is the periodicity rank of Definition 3.4. "Cell-only" keeps only edges with $t = 0$; "collapsed" keeps all edges but forgets t and merges parallel edges (node degrees shown as the set of values over node blocks).

		blocks				node c.n.		node degree, wrong graphs			
Structure (net)	atoms	$\lvert V\rvert$	nodes	linkers	$\lvert E\rvert$	computed	published	cell-only	collapsed	β_1	ϱ
HKUST-1 (tbo)	156	14	6	8	24	4	4	$\{0,2\}$	4	11	3
MOF-5 (pcu)	424	32	8	24	48	6	6	$\{3,5\}$	6	17	3
ZIF-8 (sod)	276	36	12	24	48	4	4	$\{2\}$	4	13	3
UiO-66 (fcu)	114	7	1	6	12	12	12	$\{1\}$	6	6	3

a fragment depends only on its atom multiset, which is constant on each copy. (4) Node blocks (respectively linker blocks) of $\tilde{B}$ are components, in $\tilde{B}$, of the set of node (linker) atoms; again such a component projects isomorphically onto a component of B with trivial net voltage, so the blocks of $\tilde{B}$ are exactly the pairs (β, c) with β a block of B and $c \in \mathbb{Z}_n^3$. (5) An edge joins blocks (β, c) and (β', c') in $K(\tilde{B})$ iff some atomic bond of $\tilde{B}$ does, and a bond (a, b) with block voltage t as in Equation (5) lifts to bonds joining (β, c) to $(\beta', c + t)$ for every c. Hence $K(\tilde{B})$ has vertex set $V \times \mathbb{Z}_n^3$ and edges $((i, j, t), c) \colon (i, c) \to (j, c + t)$, which is the derived graph G_n of Definition 3.1. □

Remark 4.2. The hypothesis excludes rod-shaped secondary building units (infinite chains, as in MIL-47-type frameworks), which have a non-trivial net voltage; for those, blocks are periodic and the coarse-graining is a separate construction. None of the benchmark structures contains one. The statement is combinatorial: it assumes bonds are the lifts of the cell's bonds. Whether the *geometric* perception of § 4.2, run afresh on the supercell cell, finds the same bonds is an empirical question, answered in Table 2.

4.6 Validation against published crystallography

In the quotient graph the degree of a node block is the number of distinct linker attachments, i.e. the crystallographic coordination number, which is published for all four benchmark nets. A free consistency identity follows from Proposition 3.3(b): metal-oxo blocks never bond to a block of the same role, so Γ is bipartite and

$$\sum_{v \in V_N} \deg v = \sum_{v \in V_L} \deg v = |E| \,. \tag{6}$$

Table 3 lists the computed quantities. The identity holds in all four cases ($6 \cdot 4 = 8 \cdot 3 = 24$, $8 \cdot 6 = 24 \cdot 2 = 48$, $12 \cdot 4 = 24 \cdot 2 = 48$, $1 \cdot 12 = 6 \cdot 2 = 12$).

All four node degrees agree with the literature, and the linker degrees are $3, 2, 2, 2$ as expected (tritopic trimesate; ditopic terephthalate and imidazolate). The table also shows why the labels

cannot be dropped: a graph restricted to the cell gets every structure wrong, and collapsing parallel edges is harmless for three structures but halves the coordination of UiO-66, where each of the six linkers attaches to the single Zr_6 cluster through two *different* translations and so contributes two edges, not one. UiO-66 is the decisive test case because its cell contains one node block.

5 Spectral theory of periodic nets and their supercells

The voltage graph (Γ, ν) describes an infinite net through a finite object. This section shows that the *spectrum* of every supercell G_n is determined by $|V| \times |V|$ matrices, and derives a size-independent quantity, the stiffness tensor Σ, from the long-wavelength limit of that spectrum. These results supply the exact mathematics behind the "$\lambda_2 \sim n^{-2}$" observation that an earlier, purely empirical stage of the project reported.

5.1 Bloch decomposition

Write $\mathcal{L}_n = \mathcal{L}(G_n)$. For $k \in \mathbb{R}^3$ define the $|V| \times |V|$ Hermitian matrix

$$\mathcal{L}(k)_{ab} \;=\; \deg(a)\,\delta_{ab} \;-\; \sum_{e\in E:\; o(e)=a,\; h(e)=b} e^{\mathrm{i}k\cdot\nu(e)} \;-\; \sum_{e\in E:\; o(e)=b,\; h(e)=a} e^{-\mathrm{i}k\cdot\nu(e)}. \tag{7}$$

Equivalently $\mathcal{L}(k) = \sum_e \beta_e(k)\beta_e(k)^*$ with $\beta_e(k) = \mathbf{e}_{o(e)} - e^{\mathrm{i}k\cdot\nu(e)}\mathbf{e}_{h(e)}$ (a loop contributes $2 - 2\cos(k\cdot\nu(e))$ to the diagonal), so $\mathcal{L}(k) \succeq 0$. Let $K_n = \frac{2\pi}{n}\mathbb{Z}^3_n$ be the dual group of $\mathbb{Z}^3_n$.

Theorem 5.1 (Bloch decomposition [proved, verified])**.** *The Laplacian of the n-supercell is unitarily equivalent to a direct sum of n^3 small matrices,*

$$\mathcal{L}_n \;\cong\; \bigoplus_{k\in K_n} \mathcal{L}(k), \qquad \operatorname{spec}\mathcal{L}_n = \biguplus_{k\in K_n} \operatorname{spec}\mathcal{L}(k) \quad (\textit{multiset union}). \tag{8}$$

Proof. Identify $\mathbb{R}^{V(G_n)} \cong \mathbb{R}^V \otimes \mathbb{R}^{\mathbb{Z}^3_n}$ and use the orthonormal characters $\chi_k(c) = n^{-3/2}e^{\mathrm{i}k\cdot c}$, $k \in K_n$, which form a basis of $\mathbb{C}^{\mathbb{Z}^3_n}$. For an edge e with voltage $t = \nu(e)$, the edge (e, c) joins $(o(e), c)$ to $(h(e), c + t)$, and therefore

$$\mathcal{L}_n = \sum_{e\in E}\sum_{c\in\mathbb{Z}^3_n} b_{(e,c)}b_{(e,c)}^{\mathsf{T}}, \qquad b_{(e,c)} = \mathbf{e}_{o(e)}\otimes\delta_c - \mathbf{e}_{h(e)}\otimes\delta_{c+t}.$$

Let S_t denote the translation $(S_t f)(c) = f(c + t)$ on $\mathbb{C}^{\mathbb{Z}^3_n}$. Then $\sum_c b_{(e,c)}b_{(e,c)}^{\mathsf{T}} = \mathbf{e}_o\mathbf{e}_o^{\mathsf{T}}\otimes I + \mathbf{e}_h\mathbf{e}_h^{\mathsf{T}}\otimes I - \mathbf{e}_o\mathbf{e}_h^{\mathsf{T}}\otimes S_t^{\mathsf{T}} - \mathbf{e}_h\mathbf{e}_o^{\mathsf{T}}\otimes S_t$. Every character is an eigenvector of the translation, $S_t\chi_k = e^{\mathrm{i}k\cdot t}\chi_k$, $S_t^{\mathsf{T}}\chi_k = e^{-\mathrm{i}k\cdot t}\chi_k$ (on a torus S_t is unitary, so $S_t^{\mathsf{T}} = S_t^{-1}$). Acting on $\varphi\otimes\chi_k$ with $\varphi \in \mathbb{C}^V$ therefore gives $\big(\mathcal{L}(k)\varphi\big)\otimes\chi_k$ with $\mathcal{L}(k)$ as in Equation (7). The subspaces $\mathbb{C}^V\otimes\chi_k$ are orthogonal, invariant and span $\mathbb{C}^{V(G_n)}$, which proves the claim. Since $\mathcal{L}_n$ is real symmetric, $\mathcal{L}(-k) = \overline{\mathcal{L}(k)}$ and so $\operatorname{spec}\mathcal{L}(k) = \operatorname{spec}\mathcal{L}(-k)$. □

Numerical verification. For $n \in \{1, 2, 3\}$ and all four benchmark frameworks we assembled $\mathcal{L}_n$ explicitly ($n^3\,|V|$ up to $27 \cdot 36 = 972$ rows) and compared its sorted spectrum against the union of the n^3 spectra of Equation (7). The maximal absolute difference was 1.7×10^{-14}, i.e. machine precision (`check_02_bloch_tail.py`).

Corollary 5.2 (Algebraic connectivity of supercells [proved])**.** *Let Γ be connected. Then*

$$\lambda_2(\mathcal{L}_n) \;=\; \min\Big\{\lambda_2\big(\mathcal{L}(0)\big),\; \min_{k\in K_n\setminus\{0\}} \lambda_1\big(\mathcal{L}(k)\big)\Big\}, \tag{9}$$

where $\lambda_1(\mathcal{L}(k)) = \min\operatorname{spec}\mathcal{L}(k)$. Moreover $\lambda_1(\mathcal{L}(k)) = 0$ for $k \neq 0$ if and only if $k\cdot\nu(W) \in 2\pi\mathbb{Z}$ for every closed walk W, i.e. if and only if k annihilates the net voltage group H; if $H = \mathbb{Z}^3$ this never happens for $k \in K_n \setminus \{0\}$.

Proof. By Theorem 5.1, 0 is an eigenvalue of $\mathcal{L}(0)$ with eigenvector $\mathbf{1}$, simple because Γ is connected (Equation (1)). The second-smallest eigenvalue of the union is therefore the smaller of $\lambda_2(\mathcal{L}(0))$ and the smallest eigenvalue over all $k \neq 0$. If $\mathcal{L}(k)\varphi = 0$ then $\sum_e \left|\varphi(o(e)) - e^{\mathrm{i}k\cdot\nu(e)}\varphi(h(e))\right|^2 = 0$, so $\varphi(o(e)) = e^{\mathrm{i}k\cdot\nu(e)}\varphi(h(e))$ for every edge. Transporting φ around a closed walk W multiplies it by $e^{\mathrm{i}k\cdot\nu(W)}$, which must equal 1 for $\varphi \neq 0$ (a nonzero φ is nowhere zero along the walk). Conversely, if k kills H, define φ by transport from a base vertex along a spanning tree; it is well-defined and lies in the kernel. □

5.2 Long-wavelength limit: the stiffness tensor Σ

Since $\mathcal{L}(k)$ is analytic in k and 0 is a simple eigenvalue of $\mathcal{L}(0)$, Kato's theory [15] provides an analytic branch $\lambda_1(\mathcal{L}(k))$ with $\lambda_1(\mathcal{L}(0)) = 0$, and because spec $\mathcal{L}(k) =$ spec $\mathcal{L}(-k)$ the branch is even, so $\lambda_1(\mathcal{L}(k)) = k^{\mathsf{T}}\Sigma k + O(|k|^4)$ for a symmetric positive semidefinite 3×3 matrix $\Sigma = \Sigma(\Gamma, \nu)$. The following theorem computes Σ in closed form.

Theorem 5.3 (Homogenised stiffness tensor [proved, verified])**.** *Let $T \in \mathbb{R}^{E\times 3}$ have rows $\nu(e)^{\mathsf{T}}$, let $B = [b_e]_{e\in E} \in \mathbb{R}^{V\times E}$ be the incidence matrix, and let $g = BT \in \mathbb{R}^{V\times 3}$, $g = \sum_e b_e \nu(e)^{\mathsf{T}}$. Then, in lattice coordinates,*

$$\boxed{\Sigma \;=\; \frac{1}{|V|}\Big[\sum_{e\in E}\nu(e)\nu(e)^{\mathsf{T}} \;-\; g^{\mathsf{T}}\mathcal{L}(\Gamma)^{+}g\Big]} \tag{10}$$

where $\mathcal{L}(\Gamma)^+$ is the Moore–Penrose pseudo-inverse of the base Laplacian. Equivalently

$$k^{\mathsf{T}}\Sigma k = \frac{1}{|V|}\min_{\chi\in\mathbb{R}^V}\left\|Tk - B^{\mathsf{T}}\chi\right\|^2, \tag{11}$$

the squared distance of the voltage form Tk from the space of coboundaries. The matrix Σ is positive semidefinite, it is positive definite iff rank $H = 3$*, and the supercell spectral gap obeys*

$$\lim_{n\to\infty} n^2\lambda_2(\mathcal{L}_n) \;=\; 4\pi^2 \min_{m\in\mathbb{Z}^3\setminus\{0\}} m^{\mathsf{T}}\Sigma\, m \qquad (\text{if } H = \mathbb{Z}^3). \tag{12}$$

Proof. Closed form. By Rayleigh–Ritz, $\lambda_1(\mathcal{L}(k)) = \min_{\varphi\neq 0}\langle\varphi, \mathcal{L}(k)\varphi\rangle / \|\varphi\|^2$ with $\langle\varphi, \mathcal{L}(k)\varphi\rangle = \sum_e \left|\varphi(o) - e^{\mathrm{i}k\cdot t_e}\varphi(h)\right|^2$. Kato's perturbation theory gives the second-order coefficient of the simple eigenvalue as the minimum, over first-order corrections of the eigenvector, of the second-order part of the Rayleigh quotient. Write $\varphi = |V|^{-1/2}(\mathbf{1} + \mathrm{i}\chi)$ with $\chi \in \mathbb{R}^V$ of order $|k|$ (a real first-order correction is the only one that matters because the zeroth-order vector is real and the quadratic form is invariant under the global phase). Expanding $e^{\mathrm{i}k\cdot t} = 1 + \mathrm{i}k\cdot t + O(|k|^2)$,

$$\varphi(o) - e^{\mathrm{i}k\cdot t}\varphi(h) = \frac{\mathrm{i}}{\sqrt{|V|}}\Big[(\chi_o - \chi_h) - k\cdot t\Big] + O(|k|^2),$$

so to second order $\langle\varphi, \mathcal{L}(k)\varphi\rangle = |V|^{-1}\sum_e \left(b_e^{\mathsf{T}}\chi - k\cdot t_e\right)^2 = |V|^{-1}\left\|B^{\mathsf{T}}\chi - Tk\right\|^2$ and $\|\varphi\|^2 = 1 + O(|k|^2)$. Minimising over χ is a least-squares problem whose solution is $\chi = \mathcal{L}^+BTk$ (using $\mathcal{L} = BB^{\mathsf{T}}$, so that the orthogonal projector onto range B^{T} is $B^{\mathsf{T}}\mathcal{L}^+B$). The minimum value is $\left\|(I - B^{\mathsf{T}}\mathcal{L}^+B)Tk\right\|^2 = k^{\mathsf{T}}\big(T^{\mathsf{T}}T - T^{\mathsf{T}}B^{\mathsf{T}}\mathcal{L}^+BT\big)k$, which is Equation (10) since $T^{\mathsf{T}}T = \sum_e t_e t_e^{\mathsf{T}}$.

Definiteness. $k^{\mathsf{T}}\Sigma k = 0$ iff $Tk = B^{\mathsf{T}}\chi$ for some χ, i.e. the real-valued 1-cochain $e \mapsto k\cdot t_e$ is a coboundary, i.e. its sum along every closed walk vanishes: $k\cdot\nu(W) = 0$ for all W. This means $k \perp H$, so $\ker\Sigma = H^{\perp}$ and $\Sigma \succ 0$ iff rank $H = 3$.

Limit Equation (12). By Corollary 5.2 (with $H = \mathbb{Z}^3$) and the expansion above, $n^2\lambda_2(\mathcal{L}_n) = \min_m n^2\lambda_1(\mathcal{L}(2\pi m/n))$ over nonzero $m \in \mathbb{Z}^3_n$ (taking representatives with $|m|_\infty \le n/2$), and $n^2\lambda_1(\mathcal{L}(2\pi m/n)) = 4\pi^2\, m^{\mathsf{T}}\Sigma m\,(1 + O(|m|^2/n^2))$. The minimum over m is attained at bounded

Table 4. Stiffness tensor of the benchmark nets. Σ_{lat} is isotropic only for the cubic case; for HKUST-1 (rhombohedral primitive cell) it has diagonal $1/14$ and off-diagonal $-1/42$. The Cartesian eigenvalues (in Å^2) are triply degenerate to the printed precision and are identical for the cell and a supercell, as predicted by Proposition 5.5.

Structure	$\lvert V\rvert$	β_1	$\Sigma_{\text{lat},11}$	eig Σ_{cart} (Å^2)	$4\pi^2\Sigma_{11}$	$n^2\lambda_2$ ($n = 16$)
HKUST-1	14	11	0.0714	16.465	2.8199	2.8153
MOF-5	32	17	0.03125	20.908	1.2337	1.2305
ZIF-8	36	13	0.01389	4.010	0.5483	0.5479

m because $\Sigma \succ 0$, which proves the limit. (The term $\lambda_2(\mathcal{L}(0))$ in Equation (9) is n-independent, hence irrelevant for large n.) □

Remark 5.4 (Meaning of Σ). Σ is the *homogenised* (effective) diffusion tensor of the continuous-time random walk with generator $-\mathcal{L}$ on the net, in the sense of Kotani and Sunada [16, 22]: since $\mathbb{E}\, e^{ik\cdot X_t} \approx e^{-\lambda_1(\mathcal{L}(k))t}$, the displacement satisfies $\mathrm{Cov}(X_t) \approx 2\Sigma t$. It depends on the net and not on how its unit cell is chosen (Proposition 5.5). It is the graph-level analogue of a pore-network diffusion tensor, but it describes hopping between coarse-grained building blocks and not diffusion of guest molecules; we make no claim about the latter.

Proposition 5.5 (Cell- and size-independence of Σ [proved, verified])**.** *Let M be the 3×3 matrix whose rows are the lattice vectors, let Σ_{lat} be the tensor Equation (10), and define the Cartesian tensor $\Sigma_{\text{cart}} = M^{\mathsf{T}}\Sigma_{\text{lat}}M$ (units Å^2). If the net voltage group is $H = \mathbb{Z}^3$, then Σ_{cart} is unchanged when the unit cell is replaced by any $m_1 \times m_2 \times m_3$ supercell (i.e. when (Γ, ν) is replaced by the voltage graph of $G_{\mathbf{m}}$ with enlarged lattice $\mathrm{diag}(\mathbf{m})M$).*

Proof. Let $\Lambda \supset \Lambda'$ be the old and the new translation lattices, with reciprocal lattices $\Lambda^* \subset \Lambda'^*$. The Bloch matrix of the enlarged cell at a small wave vector κ is unitarily equivalent to $\bigoplus_{\eta \in \Lambda'^*/\Lambda^*} \mathcal{L}(\kappa + \eta)$ ("band folding", the same computation as Theorem 5.1 carried out over a finite-index sublattice). For $\eta \neq 0$ the smallest eigenvalue of $\mathcal{L}(\kappa + \eta)$ is bounded below by a positive constant for $|\kappa|$ small, since $\lambda_1(\mathcal{L}(\eta)) > 0$ by Corollary 5.2 (this is where $H = \mathbb{Z}^3$ is used). Hence the lowest branch of the enlarged cell near $\kappa = 0$ is the lowest branch of the original cell, i.e. the function $\kappa \mapsto \kappa^{\mathsf{T}}\Sigma_{\text{cart}}\kappa + O(|\kappa|^4)$ is unchanged (in Cartesian wave-vector coordinates $\kappa_{\text{cart}} = M^{-1}\kappa_{\text{lat}}$). □

Numerical verification (Table 4). For the three cubic-type frameworks the tensor is isotropic, and it is the *same* to 10^{-13} when evaluated on the unit cell and on a $2 \times 2 \times 2$ supercell. For MOF-5 we obtain $\Sigma_{\text{lat}} = \frac{1}{32} I$ exactly, hence $4\pi^2\Sigma_{\text{lat},11} = 1.2337$, which matches the extrapolated $n^2\lambda_2(\mathcal{L}_n) \to 1.2305$ at $n = 16$ (the discrepancy of 0.3% is the expected $O(n^{-2})$ term). The quadratic coefficient of $\lambda_1(\mathcal{L}(k))$ measured at $k \approx 10^{-3}$ agrees with Equation (10) to a relative error of order 10^{-6}, consistent with the $O(|k|^2)$ correction (`check_06_stiffness.py`).

5.3 Consequences for the graph representation: symmetry degeneracy and defects

A benefit of the periodic formulation is that the *orbits* of the crystal symmetry group are visible in every graph invariant. In a defect-free framework all node blocks are equivalent under the space group, as are all linker blocks, so any invariant defined by the graph alone takes at most as many values as there are vertex orbits. Table 5 shows this for the four benchmarks: betweenness and eigenvector centrality take *exactly two* distinct values on the perfect $2 \times 2 \times 2$ supercell (one for nodes, one for linkers). Deleting a single linker destroys the symmetry and splits these into many classes (e.g. 93 for ZIF-8).

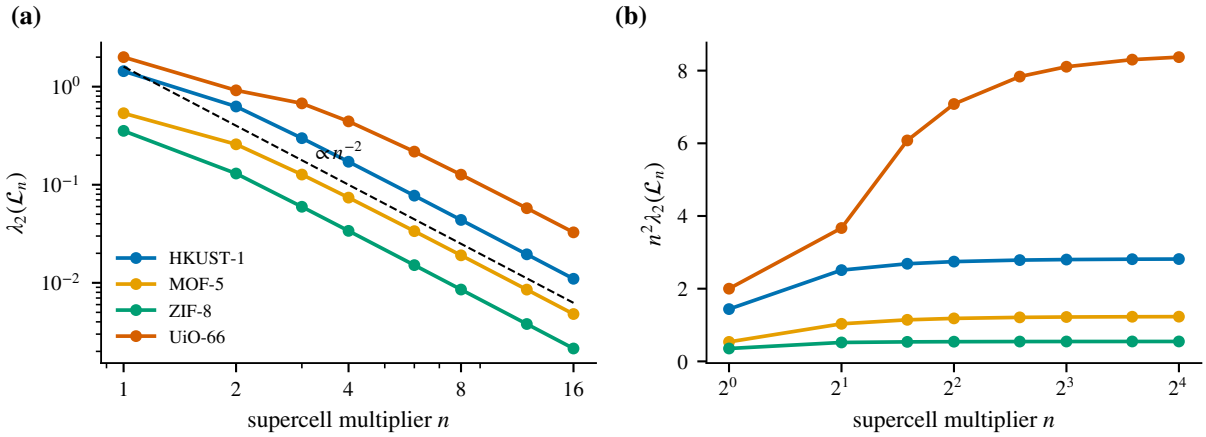


Figure 3. Supercell spectral gap. (a) $\lambda_2(\mathcal{L}_n)$ against the supercell multiplier n on logarithmic axes: all frameworks follow the predicted slope -2. (b) The rescaled gap $n^2\lambda_2(\mathcal{L}_n)$ converges to the constant $4\pi^2 \min_m m^\mathsf{T}\Sigma m$ of Theorem 5.3 from below, with an $O(n^{-2})$ correction.

Table 5. Number of distinct values of betweenness (BC) and eigenvector (EC) centrality on the $2 \times 2 \times 2$ supercell graph. "Defect" means one linker vertex deleted. Also listed: the relative drop of the algebraic connectivity caused by that defect, for the cell quotient and for the supercell.

	BC classes		EC classes		$\Delta\lambda_2/\lambda_2$	
Structure	perfect	defect	perfect	defect	cell	supercell
HKUST-1	2	20	2	15	-17.5%	-12.3%
MOF-5	2	16	2	16	-13.0%	-4.7%
ZIF-8	2	93	2	78	-15.4%	-7.2%
UiO-66	2	9	2	9	0.0%	-5.8%

Two lessons follow. First, distinct centrality values in an experimentally determined structure are a *graph-level defect detector*: a perfect net cannot produce more classes than it has vertex orbits (this is a consequence of the invariance of betweenness and of the Perron vector under graph automorphisms, together with the fact that the space group acts by automorphisms). Second, the effect of a defect on λ_2 *depends on the size of the cell in which it is embedded*: the same single vacancy lowers λ_2 by 17.5% in the HKUST-1 unit cell but by 12.3% in a $2 \times 2 \times 2$ supercell, because in the small cell the vacancy is repeated periodically and so has the character of a *defect lattice*, not an isolated defect. For UiO-66 the unit-cell value is unchanged ($\lambda_2 = 2$ before and after) while the supercell value drops by 5.8%; here the one-node cell quotient is too small to register the vacancy. This is a further instance of the finite-size principle that Theorem 5.3 makes exact.

5.4 Interlacing bounds for defects

Proposition 3.6 gives a rigorous, size-free bound for the change in *every* eigenvalue caused by removing s edges: $\lambda_k(\mathcal{L}') \in [\lambda_{k-s}, \lambda_k]$. Removing one linker of degree d therefore shifts λ_2 by at most $d + 1$ (an upper bound on the effect of any single vacancy, attained only for small graphs); for the supercell the observed shifts of 0.05–0.08 (absolute) are far inside this bound, which shows that the bound is useful for *ruling out* large effects but is not sharp for large supercells. A sharp result requires the localisation theory of Theorem 5.3 and is left open.

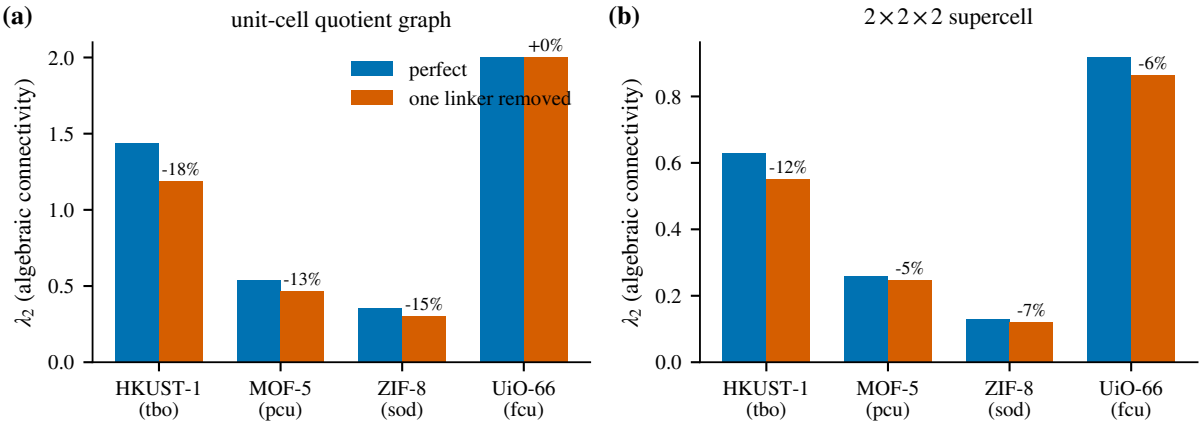


Figure 4. Effect of removing one linker on the algebraic connectivity λ_2. (a) Unit-cell quotient graph; (b) $2 \times 2 \times 2$ supercell. The percentages are relative changes. A defect in a small cell is effectively a defect *lattice*.

6 Multifractal analysis of the coarse-grained graph

The first analysis stage of the project followed the node-based multifractal analysis (NMFA/iNMFA) of Xiao, Wang and Li [25], applied to the block graph of Section 4. We formalise it here precisely enough to prove what the estimator can and cannot detect. The outcome is negative for our original hypothesis and is reported as such: the width of the multifractal spectrum is governed by the number of vertices, not by the chemistry of the framework (Theorem 6.3).

6.1 Definitions

Let $G = G_n$ be a supercell graph with $N = n^3\,|V|$ vertices and graph diameter D. Let $d(u, v)$ be the shortest-path distance.

Definition 6.1 (Box covering, measure, partition function)**.** For an integer radius $r \in \{1, \dots, D\}$ a *box covering* $\mathcal{B}_r$ is a partition of $V(G)$ into sets B of diameter $< r$ built by the greedy random procedure of Song, Havlin and Makse [20]: pick an uncovered vertex uniformly at random, put into its box all uncovered vertices within distance $< r$, repeat. Let $B_r(i)$ denote the box that contains vertex i. Let $I \subseteq V$ be a fixed set of *influential* vertices (the top 10% by degree in the iNMFA variant, all vertices in NMFA). The *probability measure* and *partition function* in trial s are

$$p_i^{(s)}(r) = \frac{\left|B_r^{(s)}(i)\right|}{N}, \qquad Z_q^{(s)}(r) = \sum_{i \in I} p_i^{(s)}(r)^q, \qquad q \in \mathbb{R}. \tag{13}$$

The *mass exponent* is defined by the log–log regression through the origin

$$\tau(q) = \frac{\sum_{k=1}^{K} x_k\, y_k(q)}{\sum_{k=1}^{K} x_k^2}, \qquad x_k = \ln \frac{r_k}{r_N}, \quad y_k(q) = \mathbb{E}_s \ln Z_q^{(s)}(r_k), \tag{14}$$

where $r_1 < \cdots < r_K = r_N = D$ are the radii used (all integers from 1 to D), so that $x_k \le 0$ and $x_K = 0$, and the average is over independent random coverings. The *singularity exponent* and *multifractal spectrum* are the Legendre transform

$$\alpha(q) = \frac{\mathrm{d}\tau}{\mathrm{d}q}, \qquad f(\alpha(q)) = q\,\alpha(q) - \tau(q), \tag{15}$$

evaluated on a grid $q \in [-10, 10]$ with spacing 0.5, derivatives by central differences. The spectral width is $\Delta\alpha = \max_q \alpha(q) - \min_q \alpha(q)$, which Xiao et al. [25] interpret as the degree of structural heterogeneity, and α_0 is the value of α at the maximum of f.

Two implementation points have mathematical content. First, the logarithm must be taken *before* averaging over trials, $y_k = \mathbb{E} \ln Z_q$ and not $\ln Z_q(\mathbb{E}p)$; averaging the measure first replaces the random measure by its mean and biases the spectrum narrow (a defect we found in the earlier code and corrected). Second, the covering is computed on the full graph while the sum in Equation (13) is restricted to I; restricting the *graph* to I would give an edgeless graph, since metal-oxo nodes never share a bond [3].

6.2 Properties of the estimator

Proposition 6.2 (Shape of the spectrum [proved, verified])**.** *Let $x_k \le 0$ not all zero. Then:*

(a) τ in Equation (14) is concave *in q, so $\alpha(q) = \tau'(q)$ is non-increasing and f is a concave function of α with $f'(\alpha) = q$ and its maximum at $q = 0$.*

(b) $\tau(0) = -\rho_D \ln |I|$ and $f(\alpha(0)) = \rho_D \ln |I|$, where

$$\rho_D := \frac{\sum_k |x_k|}{\sum_k x_k^2}. \tag{16}$$

In particular the peak of the spectrum, which in the fractal literature is identified with the capacity dimension D_0, equals $\rho_D \ln |I|$ and is not *a dimension unless $\rho_D \ln |I|$ happens to coincide with one.*

(c) If instead the regression of y_k on x_k has a free intercept, then $\tau_{\text{free}}(0) = 0$ for every graph and every covering (because $y_k(0) = \ln |I|$ is constant), and $f(\alpha(0)) = 0$.

Proof. (a) For fixed s and k, $q \mapsto \ln Z_q^{(s)}(r_k) = \ln \sum_{i \in I} e^{q \ln p_i}$ is a log-sum-exp function of q, hence convex; expectations and non-negative combinations of convex functions are convex, so y_k is convex. Since $x_k \le 0$, $\tau = \sum_k (x_k / \sum x^2)\, y_k$ is a non-positive combination of convex functions and therefore concave. Then $\alpha = \tau'$ is non-increasing; the Legendre transform of a concave function is concave, with $\mathrm{d}f/\mathrm{d}\alpha = q$, so f is maximal where $q = 0$. (b) At $q = 0$, $Z_0 = |I|$ for every covering, so $y_k(0) = \ln |I|$ and $\tau(0) = \ln |I| \sum_k x_k / \sum_k x_k^2 = -\rho_D \ln |I|$; also $f(\alpha(0)) = 0 \cdot \alpha - \tau(0)$. (c) A constant regressand has slope zero. □

Verification. On HKUST-1 ($n = 3$, $N = 378$, $|I| = 76$, i.e. the top 20% by degree in this demonstration) the through-origin fit gives $\tau(0) = -\rho_D \ln 76 = -0.686 \cdot 4.331 = -2.97$, $f_{\max} = 2.97$ and $q_{\text{peak}} = 0$; the free-intercept fit gives $\tau(0) = 0$ to machine precision (Figure 5). The apparent "fractal dimension" D_0 read from a through-origin spectrum is therefore an artefact of the number of scales. The original project code had reported this peak without this caveat.

6.3 The tails of the spectrum are fixed by the size of the graph

The width $\Delta\alpha$ is controlled by the two ends $q \to \pm\infty$ of the spectrum, where the partition function is dominated by the rarest and the most common boxes.

Theorem 6.3 (Tail identity [proved, verified])**.** *Let*

$$p_{\min}^{(s)}(r) = \min_{i \in I} p_i^{(s)}(r), \qquad p_{\max}^{(s)}(r) = \max_{i \in I} p_i^{(s)}(r),$$

and let $m_k = N\, p_{\min}(r_k) \in \{1, \dots, N\}$ be the size in vertices of the smallest box that contains an influential vertex. Then the limits $\alpha(\pm\infty) = \lim_{q \to \pm\infty} \alpha(q)$ exist and

$$\alpha(-\infty) = \frac{\sum_k x_k\, \mathbb{E} \ln p_{\min}(r_k)}{\sum_k x_k^2} \;=\; \rho_D \ln N \;-\; \mu_D, \quad \mu_D := \frac{\sum_k |x_k|\; \mathbb{E} \ln m_k}{\sum_k x_k^2} \in \big[0, \rho_D \ln N\big], \tag{17}$$

$$\alpha(+\infty) = \frac{\sum_k x_k\, \mathbb{E} \ln p_{\max}(r_k)}{\sum_k x_k^2} \;\ge\; 0. \tag{18}$$

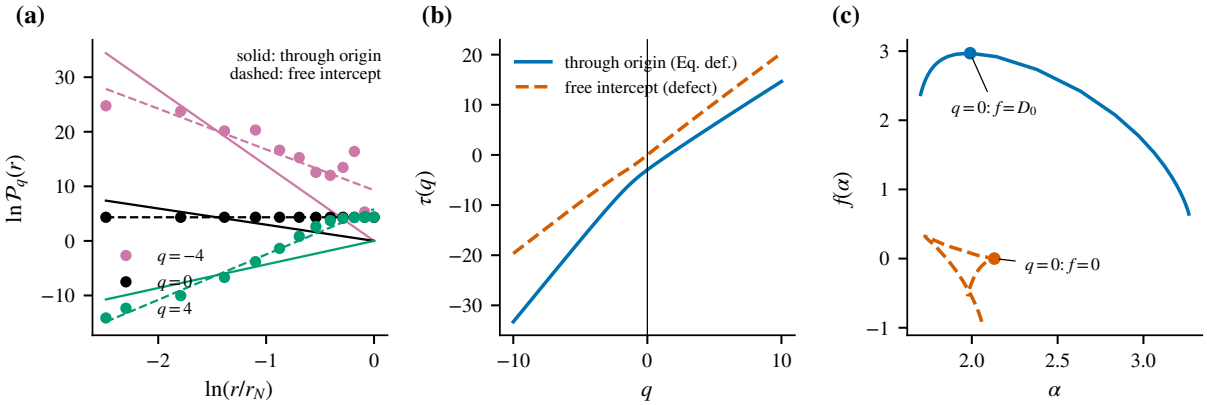


Figure 5. Anatomy of the estimator on HKUST-1 ($n = 3$). (a) $\ln Z_q(r)$ against $\ln(r/r_N)$ for selected q; the last point ($x = 0$) has $Z_q = \text{const}$, which is why Equation (14) is forced through the origin. (b) $\tau(q)$ from the through-origin fit (blue) is concave, with $\tau(0) = -\rho_D \ln|I| \neq 0$; the free-intercept fit (red, dashed) has $\tau(0) = 0$. (c) The spectrum $f(\alpha)$ peaks at $q = 0$ with height $\rho_D \ln|I|$, in agreement with Proposition 6.2.

Consequently the total width obeys

$$\Delta\alpha \;=\; \alpha(-\infty) - \alpha(+\infty) \;=\; \rho_D \ln N - \mu_D - \alpha(+\infty) \qquad \textit{(untruncated, } q \in \mathbb{R}\textit{)}, \tag{19}$$

and for the truncated range $q \in [-q_0, q_0]$ the width differs from Equation (19) by an amount that tends to 0 exponentially in q_0, at a rate set by the gap between the two smallest (resp. two largest) distinct values of p_i.

Proof. By Equation (14), $\alpha(q) = \sum_k x_k\, \partial_q y_k(q) / \sum_k x_k^2$, and $\partial_q \ln Z_q = \sum_i p_i^q \ln p_i / \sum_i p_i^q$, the mean of $\ln p_i$ under the tilted measure $\propto p_i^q$. As $q \to -\infty$ this measure concentrates on the set where p_i is minimal, so $\partial_q \ln Z_q \to \ln p_{\min}$, with error $O\big(|I|\,(p_{\min}/p')^{|q|}\big)$, p' the next larger value; dominated convergence over the finite set of trials and scales then gives the first expression in Equation (17). Writing $\ln p_{\min} = -\ln N + \ln m_k$ and using $\sum_k x_k = -\sum_k |x_k|$ yields $\rho_D \ln N - \mu_D$. Since $1 \le m_k \le N$, $0 \le \mu_D \le \rho_D \ln N$. The case $q \to +\infty$ is identical with $p_{\max} \le 1$, which gives $\ln p_{\max} \le 0$ and, since $x_k \le 0$, $\alpha(+\infty) \ge 0$. The truncation statement follows from the stated error rate. □

Corollary 6.4 (Width is not a material property [proved])**.** *For any family of graphs G for which μ_D and $\alpha(+\infty)$ stay bounded, $\Delta\alpha = \rho_D \ln N + O(1)$ diverges logarithmically with the number of vertices. For nets of dimension three, with integer radii $1, \dots, D$, $x_k = \ln(k/D)$ and $\rho_D \to \frac{1}{2}$ (because $\sum |x_k| \approx D \int_0^1 (-\ln u)\mathrm{d}u = D$ and $\sum x_k^2 \approx D \int_0^1 \ln^2 u\, \mathrm{d}u = 2D$), so $\Delta\alpha \approx \frac{1}{2} \ln N$ asymptotically. Hence the width can separate only structures of different* size, *not of different chemistry or topology.*

Table 6 verifies the identity on real structures, and Figure 6 displays it: the lower branch $\alpha(-10)$ follows the exact tail and the dotted curve $\rho_D \ln N$ at a bounded vertical offset $\mu_D \approx 0.8$, whereas the centre α_0 and upper tail $\alpha(+10)$ rise slowly; all three distinct quantities increase with $\ln D$. The consequence for the original project is stated in Section 7: the two "bands" of spectral width that had been reported for two families of structures were a finite-size effect and are withdrawn.

7 Computational results

All numbers below were recomputed from the CIF files for this report; Appendix A lists the script that produces each one. Structures are the four benchmarks of Table 3, plus two larger collections

Table 6. Verification of Theorem 6.3 on supercells of three benchmark frameworks, $q \in [-10, 10]$. $\alpha(-10)$ is the value on the grid, $\alpha(-\infty)$ the exact limit from Equation (17) (they agree to better than 0.011), $\rho_D \ln N - \mu_D$ reproduces $\alpha(-\infty)$ identically by construction. μ_D stays in $[0.54, 1.04]$ over a factor 27 in N while $\alpha(-\infty)$ rises by 1.0 to 1.5. The width $\Delta\alpha$ therefore grows with N for every structure.

Structure	n	N	D	$\alpha(-10)$	$\alpha(-\infty)$	$\rho_D \ln N$	μ_D	$\alpha(+10)$	$\Delta\alpha$
HKUST-1	2	112	8	2.649	2.655	3.553	0.897	1.447	1.20
	3	378	12	3.264	3.270	4.069	0.799	1.702	1.56
	4	896	16	3.653	3.658	4.420	0.762	1.843	1.81
	6	3024	24	4.117	4.120	4.906	0.786	1.992	2.12
MOF-5	2	256	12	2.769	2.778	3.802	1.024	1.402	1.37
	3	864	18	3.310	3.317	4.313	0.996	1.610	1.70
	4	2048	24	3.636	3.644	4.667	1.024	1.739	1.90
	6	6912	36	4.122	4.127	5.164	1.037	1.875	2.25
ZIF-8	2	288	12	3.140	3.150	3.883	0.733	1.805	1.34
	3	972	18	3.745	3.751	4.388	0.637	2.056	1.69
	4	2304	24	4.187	4.198	4.740	0.542	2.200	1.99

Table 7. Spectral quantities at the supercell size listed (20% influential vertices, 20 random coverings, fit window restricted to the upper half of the scales, as in `check_05_tau_spectra.py`). α_0 is the abscissa of the peak of f and $f_{\max}$ the peak height.

Structure	net	n	N	D	$\alpha_{\min}$	α_0	$\alpha_{\max}$	$f_{\max}$
HKUST-1	tbo	4	896	16	1.899	2.171	3.441	2.974
MOF-5	pcu	3	864	18	1.658	2.027	3.145	2.893
ZIF-8	sod	3	972	18	2.071	2.352	3.472	2.960
UiO-66	fcu	6	1512	12	1.993	2.581	4.043	3.459

used only for the statistical tests of Section 7.3: 72 hypothetical MOFs from hMOF [2, 24] and 55 experimental structures from CoRE MOF [5] (those of 77 and 61 inputs that completed the pipeline).

7.1 Validation of the construction

The voltage-graph pipeline reproduces the published coordination numbers of all four benchmarks (Table 3); the handshake identity Equation (6) holds in each case; decomposing first and replicating second gives a graph isomorphic to replicating first and decomposing second in 5/5 tests (Table 2), as Theorem 4.1 predicts; and the Bloch decomposition matches the explicit supercell spectrum to 1.7×10^{-14} (Theorem 5.1). The periodic-net formalism also shows, for the first time in this project, the dependence of the Laplacian gap on supercell size: $\lambda_2(\mathcal{L}_n) \sim n^{-2}$ with the constant given in closed form by Theorem 5.3.

7.2 Multifractal spectra of the four benchmarks

Figure 7 shows the spectra $f(\alpha)$ and mass exponents $\tau(q)$ of the four frameworks, and Table 7 the derived quantities. All spectra are concave with peak at $q = 0$, as Proposition 6.2 demands; τ is concave and increasing ($\alpha > 0$). Different nets give different spectra (the order of α_0 is pcu $<$ tbo $<$ sod $<$ fcu), but every number here depends on the supercell size (Section 6); the table is therefore a description of the *graphs analysed*, not of the materials.

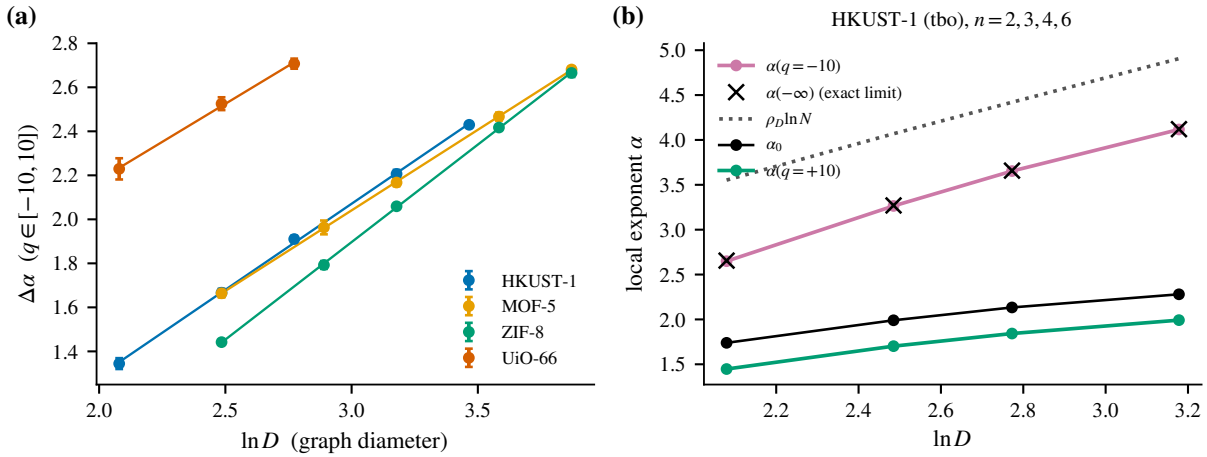


Figure 6. Finite-size behaviour of the spectrum. (a) $\Delta\alpha$ against $\ln D$ for supercells of four frameworks, with error bars over random coverings; in each case the dependence is linear in $\ln D$ with slopes between 0.69 and 0.89 ($R^2 > 0.998$). (b) For HKUST-1 ($n = 2,3,4,6$): $\alpha(-10)$, the exact limit $\alpha(-\infty)$ from Theorem 6.3 (crosses), and the asymptote $\rho_D \ln N$ (dotted) which differs from it by the bounded μ_D; α_0 and $\alpha(+10)$ grow more slowly.

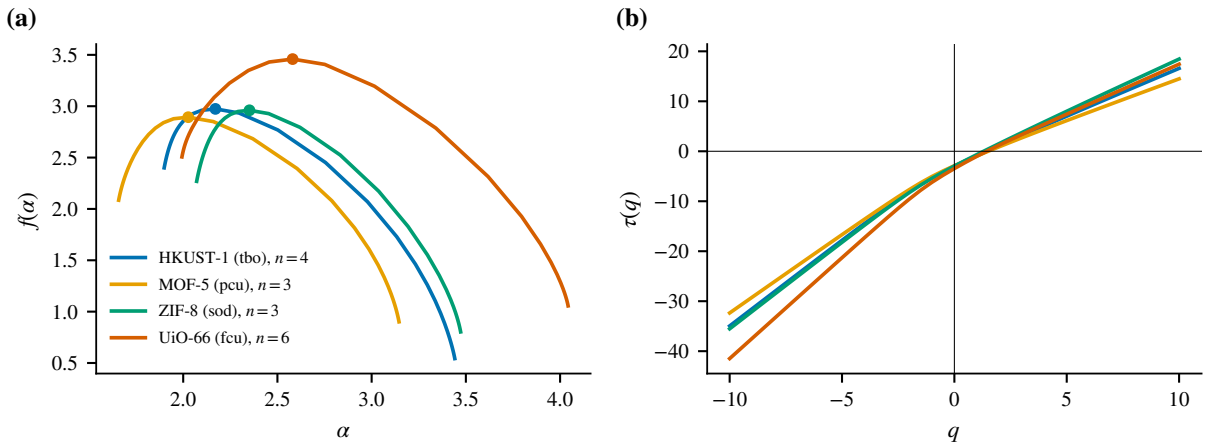


Figure 7. (a) Multifractal spectra $f(\alpha)$ and (b) mass exponents $\tau(q)$ of the coarse-grained graphs of the four benchmark frameworks. All curves are concave; the peak of each spectrum sits at $q = 0$ (Proposition 6.2).

7.3 Testing the width hypothesis on a large sample

The original project hypothesis was that the spectral width $\Delta\alpha$ of the coarse-grained graph distinguishes families of MOFs. Earlier stages of the work reported narrow bands of $\Delta\alpha$ for two families of structures (for example 1.17–1.24). Theorem 6.3 predicts that a narrow band can occur only if the structures have nearly the same number of vertices. We therefore tested the hypothesis against N directly.

Table 8 and Figure 8 show that in both samples $\Delta\alpha$ correlates strongly with $\ln N$ ($r = 0.75$ and 0.81). In the hMOF sample 65 of 72 structures have the *same* node–linker degree signature $(6,2)$, i.e. six-connected nodes and ditopic linkers, pcu-like graphs that differ chiefly in size; their width therefore clusters around a median of 1.25, which is precisely the narrow "band" that had been reported, now explained as a coincidence of size and topology. We accordingly **withdraw both bands** as evidence of any material property. A narrower conclusion survives: among structures of the *same* topology and size, differences of $\Delta\alpha$ report graph-level defects (for example a missing linker splits the centrality classes of Table 5), but this has not been tested at scale.

Table 8. Spectral width against the coarse-grained vertex count N in the two large samples. r is Pearson's correlation coefficient and ρ_s Spearman's rank correlation. The last two columns give the fraction of samples with a single (node-degree, linker-degree) signature and the number of structures whose base graph has more than one connected component.

Sample	n	$\Delta\alpha$ range	median	$r(\Delta\alpha, N)$	$r(\Delta\alpha, \ln N)$	ρ_s	multi-comp.
hMOF	72	1.12–1.72	1.25	+0.69	+0.75	+0.61	3
CoRE MOF	55	0.89–2.54	1.51	+0.72	+0.81	+0.80	3

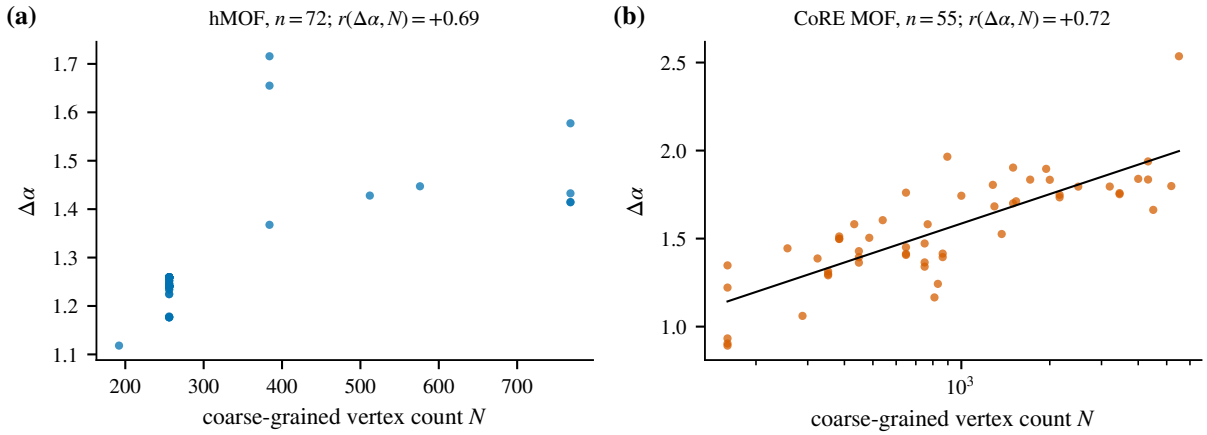


Figure 8. Spectral width against the number of coarse-grained vertices. (a) hMOF sample; (b) CoRE MOF sample (logarithmic horizontal axis, solid line: least-squares fit of $\Delta\alpha$ on $\ln N$). In both samples the correlation with N is strong, consistent with Corollary 6.4.

Remark 7.1 (Multi-component base graphs). Six structures (3 hMOF, 3 CoRE) have a base graph that is not connected, so that their net consists of several interpenetrating or layered sub-nets (e.g. $n_{\text{comp}} = 9$ in one CoRE entry). For these, Proposition 3.5 shows that ϱ and $\Delta\alpha$ should be computed per component; the distinction is carried forward in Section 8.

7.4 Finite-size trend and the two-regime scaling of the atomic graph

Figure 6 and Table 6 establish the finite-size dependence for supercells of single frameworks, and the stored slope sweeps over the two large samples (mean of the per-structure regression slopes of $\Delta\alpha$ on $\ln D$: 0.59 ± 0.13 for hMOF, 0.87 ± 0.33 for CoRE MOF, Figure 9) are consistent with the same logarithmic growth across chemistries. The slopes cluster tightly for the pcu-dominated hMOF sample and scatter widely for the heterogeneous CoRE sample, which again reflects topology and not any single material parameter.

A separate early analysis worked on the *atom-level* graph of supercells (7 296 to 27 136 atoms) and fitted the growth of ball mass with radius. Table 9 shows that a single power law fits poorly and that a two-segment fit with breakpoint at the linker diameter reduces the residual by 27–85%: a *molecular regime* with slope ≈ 1.4 up to the linker diameter, and a *framework regime* with slope 2.4–2.9 beyond it. This crossover is expected on general grounds: inside one linker the atom graph is quasi-one-dimensional and outside it fills three-dimensional space with increasing effective dimension. The breakpoint occurs at the linker length, which is set by the chemistry; we have tested this on four frameworks only, have not examined its dependence on supercell size, and treat it as a descriptive observation.

Table 9. Two-regime fit of the atom-level graph growth curve (log–log slope of ball mass against radius) for $4 \times 4 \times 4$ supercells. r_0 is the breakpoint (the linker diameter in graph steps), s_{mol} and s_{fw} are the slopes below and above it, s_1 is the slope of a single power-law fit, and "improvement" is the relative reduction of the residual sum of squares of the two-segment fit over the single fit.

Structure	atoms	r_0	s_{mol}	s_{fw}	s_1	$s_{\mathrm{fw}} - s_{\mathrm{mol}}$	improvement
HKUST-1	9 984	6	1.41	2.56	2.33	1.15	48%
MOF-5	27 136	7	1.39	2.41	2.30	1.02	27%
ZIF-8	17 664	5	1.39	2.90	2.62	1.51	85%
UiO-66	7 296	5	1.85	2.66	2.41	0.81	43%

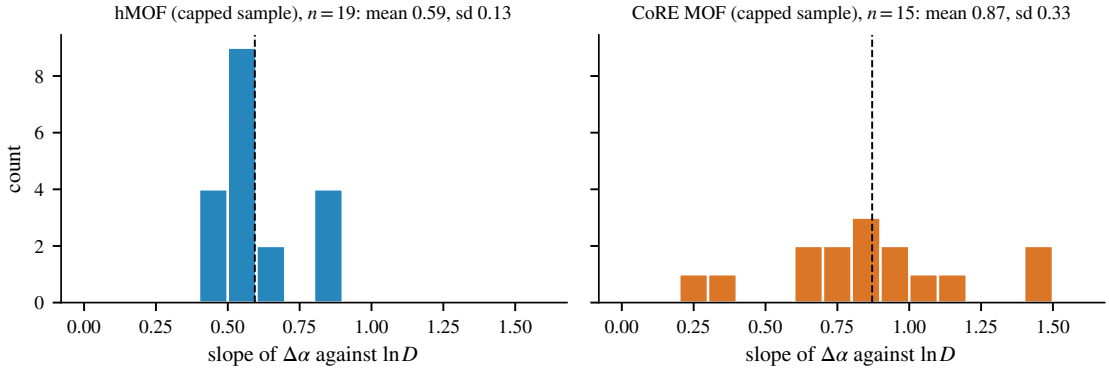


Figure 9. Distribution over structures of the slope of $\Delta\alpha$ against $\ln D$ for the two samples (dashed line: mean). Each structure contributes the slope of a regression over its own supercell sweep (two to three sizes).

7.5 Code audit

A large part of the project's effort went into a line-by-line audit of the original pipeline. Table 10 lists the defects found; each one changes a number that had been quoted earlier, and all are corrected in the code used for this paper.

8 Morphisms of frameworks: a mathematical and computational model

The review panel asked for a mathematical and computational demonstration that a framework can be morphed, for instance from a tetrahedral to a spherical or to a rod-like arrangement, since "this is known chemically but has not been shown mathematically or computationally". The literature uses the word "shape" at several length scales (a metal-oxo *cluster*, a *cage*, the *net*, a crystal *particle*). We treat two of them rigorously here and state the limit of what they establish.

M1. Dimensional morphism (topology of the net): deleting linkers takes a three-periodic net to two-, one- and zero-periodic fragments. Theorem 8.3 proves that this is a well-defined filtration of the periodicity rank ϱ with explicit component counts and spectral fingerprints.

M2. Shape morphism (geometry of a building block): the point configuration of a node is a point of a shape space with computable invariants. Propositions 8.6 and 8.7 give closed-form paths from a tetrahedron to a rod and prove that a specific pair of invariants separates tetrahedral, spherical, octahedral and rod-like configurations.

What this section does *not* show is that a synthesised crystal can be driven along such a path (that is a statement about kinetics and thermodynamics, for which the present graph model has no energy function), nor anything about adsorption. Section 9 states how the chemical-property layer

Table 10. Defects found in the project code and their consequences.

Defect	Mechanism	Consequence
Measure restricted to influential subgraph	The induced subgraph of metal nodes has no edges, since metal-oxo blocks never bond to each other	Box covering degenerates. Fix: cover the *full* graph, read the measure only on I.
Probability averaged before exponentiation	$\ln Z_q(\mathbb{E}p) \neq \mathbb{E}\ln Z_q(p)$ by Jensen's inequality	Spectrum biased narrow. Fix: average $\ln Z_q$ over trials.
Tied degrees at the 10% cutoff	Top-k selection is arbitrary among equal degrees	Influential set ambiguous. Fix: an `ambiguous` flag is reported with each run.
Through-origin R^2	Residuals of a through-origin fit scored against variance about the mean give negative R^2 even on linear data	A reported "mean $R^2 = -1.17$" was the wrong statistic. Fix: free-intercept fit as linearity diagnostic.
Wrong graph: cell-only or collapsed	Cell-only edges miss periodic bonds; collapsing parallel edges halves UiO-66's coordination	Wrong degrees and spectra (Table 3). Fix: labelled quotient graph.
Width interpreted as material property	$\Delta\alpha = \rho_D \ln N - \mu_D - \alpha(+\infty)$ (Theorem 6.3)	Bands withdrawn.

would be attached.

8.1 Dimensional morphism: the periodicity ladder

Definition 8.1 (Linker-deletion morphism)**.** Let (Γ, ν) be a voltage graph whose vertex set is partitioned into node blocks and linker blocks. For a set R of linker vertices, $\Gamma - R$ is obtained by deleting R together with all edges at R; the voltages are restricted. We call $(\Gamma, \nu) \rightsquigarrow (\Gamma - R, \nu|)$ a *linker-deletion morphism*. Write $\varrho(\Gamma - R)$ for the vector of component ranks (Definition 3.4).

Proposition 8.2 (Monotonicity of the rank [proved])**.** *If $R \subseteq R'$ then every component C' of $\Gamma - R'$ lies in a component C of $\Gamma - R$ and $H(C') \leq H(C)$; consequently $\varrho(C') \leq \varrho(C)$, and* max ϱ *is non-increasing along any chain of deletions. Moreover $\varrho(C) = 3$ is the only rank for which the derived graph has a single orbit of periodic translations, and $\varrho(C) = 0$ iff C' is finite in the derived graph.*

Proof. A closed walk in C' is a closed walk in C, so $H(C') \leq H(C)$ as subgroups of $\mathbb{Z}^3$, and rank is monotone for subgroups of a free abelian group. □

The next statement is a complete and explicit example in which all ranks $3, 2, 1, 0$ occur; it is the mathematical content of "morphing" from a three-dimensional net to lower-dimensional assemblies.

Theorem 8.3 (Periodicity ladder of MOF-5 [proved, verified])**.** *The labelled quotient graph of MOF-5 (Table 3:* 8 *nodes,* 24 *linkers) is, up to a gauge change of the voltages (a relabelling of the cell representatives that does not change the derived graph), the following. Its node vertices are N_ε with $\varepsilon \in \{0,1\}^3$ and, for every ε and axis $j \in \{1,2,3\}$, there is one linker vertex $L_{\varepsilon,j}$ joined to N_ε with voltage* 0 *and to $N_{\varepsilon+\mathbf{e}_j}$ (addition mod* 2*) with voltage $\varepsilon_j \mathbf{e}_j$. Let $S \subseteq \{1,2,3\}$ be the set of axes whose linkers are* kept*, and let R_S be the set of all linkers of the other axes. Then $\Gamma - R_S$ has*

$$2^{3-|S|} \text{ connected components, each of rank } \varrho = |S|\,, \text{ with } 2^{|S|} \text{ nodes and } |S|\,2^{|S|} \text{ linkers,} \tag{20}$$

and the total cyclomatic number is $8|S| - 8 + 2^{3-|S|}$. *For every supercell multiplier* n *the Laplacian of* $G_n(\Gamma - R_S)$ *has the eigenvalue* 0 *with multiplicity exactly*

$$\dim\ker \mathcal{L}_n = (2n)^{3-|S|}. \tag{21}$$

Proof. Structure. Two nodes are joined through a linker of axis j iff they differ only in ε_j. After deleting the linkers of the axes not in S, adjacency can change only the coordinates ε_j, $j \in S$, so the connected components are the classes of fixed $\varepsilon_{S^c} \in \{0,1\}^{3-|S|}$, which gives $2^{3-|S|}$ components, each containing $2^{|S|}$ nodes and $|S|\ 2^{|S|}$ linkers.

Rank. Every edge of a component carries a voltage in $\{0, \mathbf{e}_j\}$ with $j \in S$, so $H \leq \mathbb{Z}^S := \bigoplus_{j\in S} \mathbb{Z}\mathbf{e}_j$ and $\varrho \leq |S|$. Conversely, for $j \in S$ and any node N_ε the closed walk $N_\varepsilon \to L_{\varepsilon,j} \to N_{\varepsilon+\mathbf{e}_j} \to L_{\varepsilon+\mathbf{e}_j,j} \to N_\varepsilon$ uses two distinct linkers, traverses two edges of voltage $\varepsilon_j\mathbf{e}_j$ and $(1-\varepsilon_j)\mathbf{e}_j$ (the other edges carry voltage 0), and so has net voltage $\varepsilon_j\mathbf{e}_j + (1-\varepsilon_j)\mathbf{e}_j = \mathbf{e}_j$. Hence $H = \mathbb{Z}^S$ and $\varrho = |S|$.

Cyclomatic number. $|V| = 8 + 8|S|$, $|E| = 16|S|$, and there are $2^{3-|S|}$ components, so $\beta_1 = |E| - |V| + c = 8|S| - 8 + 2^{3-|S|}$.

Kernel. By Proposition 3.5 applied to each component with the voltages reduced modulo n, the n-supercell of a component with $H = \mathbb{Z}^S$ splits into $\left|\mathbb{Z}^3_n/\mathbb{Z}^S_n\right| = n^{3-|S|}$ components. The kernel of a graph Laplacian has one dimension per component, so $\dim\ker \mathcal{L}_n = 2^{3-|S|}n^{3-|S|}$. □

Remark 8.4 (What was verified). The structural hypothesis of Theorem 8.3 (each linker joins two nodes displaced along one lattice axis, 8 linkers per axis) was checked on the real CIF: the classification by geometric displacement gave exactly $8+8+8$ linkers. The ranks, component counts and cyclomatic numbers computed *independently* by union-find and an integer Smith-form computation on the real data are

stage ($\lvert S\rvert$)	components	ranks	sizes	β_1 (per component)
3D framework (3)	1	(3)	(32)	(17)
2D layers (2)	2	(2, 2)	(12, 12)	(5, 5)
1D chains (1)	4	(1, 1, 1, 1)	(4, 4, 4, 4)	(1, 1, 1, 1)
0D fragments (0)	8	(0^8)	(1^8)	(0^8)

in full agreement with Equation (20). The multiplicity Equation (21) for $n = 2$ was confirmed on the assembled supercell Laplacians, which have 1, 4, 16 and 64 zero eigenvalues for $|S| = 3, 2, 1, 0$, i.e. $4^{3-|S|}$ (`check_03_morphism_ladder.py`). Finally, the 200-step deletion experiment of Proposition 3.6 produced no interlacing violation.

Corollary 8.5 (Spectral signature of the morphism [proved for $|S| = 1$, verified for all]). *For* $n = 2$, *the smallest* non-zero *Laplacian eigenvalue of* $G_2(\Gamma - R_S)$ *is* 0.2583 *for* $|S| = 3$, 0.3542 *for* $|S| = 2$ *and* $2 - \sqrt{2} = 0.5858$ *for* $|S| = 1$; *there is none for* $|S| = 0$ *(all eigenvalues vanish). The* $|S| = 1$ *value is proved: each component of* $\Gamma - R_S$ *is a* 4*-cycle with net voltage* $\mathbf{e}_j$ *and its* 2*-fold lift is the cycle* C_8, *whose smallest non-zero Laplacian eigenvalue is* $2 - 2\cos(2\pi/8) = 2 - \sqrt{2}$.

Each stage of the ladder therefore has a distinct, computable invariant pair $(\varrho, \dim\ker \mathcal{L}_n)$ and a distinct first gap; Figure 10 shows the four stages of the MOF-5 $2 \times 2 \times 2$ supercell, constructed from the real CIF coordinates.

8.2 Shape morphism of a building block

Shape descriptors. Let $P = \{x_1, \ldots, x_m\} \subset \mathbb{R}^3$ be the positions of the metal atoms of a node, with centroid $\bar{x}$ and $y_i = x_i - \bar{x}$. The *gyration tensor* $\mathsf{S} = \frac{1}{m}\sum_i y_i y_i^{\mathsf{T}}$ has eigenvalues $\mu_1 \geq \mu_2 \geq \mu_3 \geq 0$

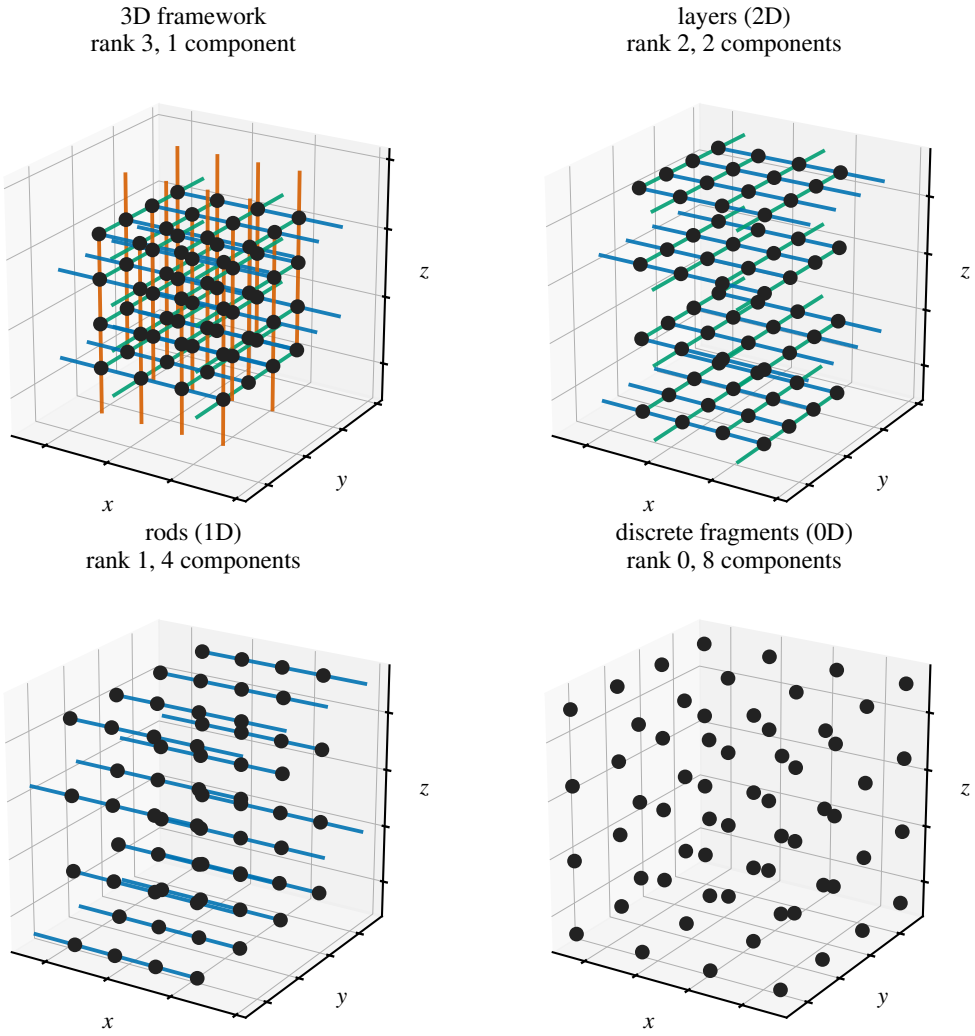


Figure 10. The periodicity ladder of MOF-5 (Theorem 8.3), drawn from the actual node and linker positions of the $2 \times 2 \times 2$ supercell. Colours code the three lattice axes of the linkers; dark dots are the Zn_4O nodes. Top left: the 3D framework (rank 3, one component). Top right: deleting the z-linkers leaves 2 layers per cell (rank 2, 2 components). Bottom left: deleting y- and z-linkers leaves 4 rods (rank 1). Bottom right: deleting all linkers leaves 8 isolated nodes (rank 0).

(units Å^2), $R_g^2 = \operatorname{tr} \mathsf{S}$, and the dimensionless *asphericity*, *acylindricity* and *relative shape anisotropy* are [23]

$$b^* = \frac{\mu_1 - \tfrac{1}{2}(\mu_2 + \mu_3)}{R_g^2}, \qquad c^* = \frac{\mu_2 - \mu_3}{R_g^2}, \qquad \kappa^2 = 1 - \frac{3(\mu_1\mu_2 + \mu_2\mu_3 + \mu_3\mu_1)}{R_g^4}. \tag{22}$$

For a sphere, cube, octahedron or regular tetrahedron $\mu_1 = \mu_2 = \mu_3$ and $b^* = c^* = \kappa^2 = 0$; for points on a line $b^* = \kappa^2 = 1$. The gyration tensor is a second-moment quantity and therefore *cannot* tell a regular tetrahedron from a sphere. The next invariant is the *traceless third moment*: with $\mathsf{M}_{abc} = \frac{1}{m}\sum_i y_{i,a} y_{i,b} y_{i,c}$ and $\mathsf{M}^0_{abc} = \mathsf{M}_{abc} - \frac{1}{5}\sum_d \left(\delta_{ab}\mathsf{M}_{cdd} + \delta_{ac}\mathsf{M}_{bdd} + \delta_{bc}\mathsf{M}_{add}\right)$, put

$$J_3^* = \frac{\left\|\mathsf{M}^0\right\|_F}{R_g^3}. \tag{23}$$

Finally, the Steinhardt bond-orientational invariants [21] of the unit vectors $\hat{y}_i$ are

$$Q_l = \Big(\frac{4\pi}{2l+1}\sum_{m=-l}^{l}\big|\bar{Y}_{lm}\big|^2\Big)^{1/2}, \qquad \bar{Y}_{lm} = \frac{1}{m}\sum_{i=1}^{m} Y_{lm}(\hat{y}_i); \tag{24}$$

they are rotation invariant and identify the point group to the extent that different groups have different non-vanishing harmonics.

A path from the tetrahedron to the rod. Take the regular tetrahedron with vertices $(1,1,1)$, $(1,-1,-1)$, $(-1,1,-1)$, $(-1,-1,1)$ and deform it by the affine map $z \mapsto sz$ along one two-fold axis, $s \in (0,\infty)$:

$$P(s) = \{(1,1,s),(1,-1,-s),(-1,1,-s),(-1,-1,s)\}. \tag{25}$$

Proposition 8.6 (Tetrahedron-to-rod path [proved, verified])**.** *Along Equation* (25), $\mathsf{S} = \mathrm{diag}(1,1,s^2)$ *and*

$$R_g^2 = 2+s^2, \qquad b^*(s) = \frac{s^2-1}{s^2+2}, \qquad \kappa^2(s) = 1 - \frac{3(2s^2+1)}{(s^2+2)^2}, \qquad J_3^*(s) = \frac{\sqrt{6}\,s}{(s^2+2)^{3/2}}. \tag{26}$$

Hence $P(1)$ is the regular tetrahedron ($b^*=0$, $J_3^* = \sqrt{2}/3$), $b^* \to 1$ and $J_3^* \to 0$ as $s \to \infty$ (a rod-like configuration: in the limit the four points form two coincident pairs on a line), and $b^* \to -\frac{1}{2}$, $J_3^* \to 0$ as $s \to 0$ (a flat square). b^* is strictly increasing in s, and J_3^* has its unique maximum at $s=1$, the regular tetrahedron, and decreases monotonically on both sides.

Proof. The tensor S is diagonal with entries $\frac{1}{4}\sum_i y_{i,a}^2 = (1,1,s^2)$ because the coordinates ± 1 average to 1 and the third coordinate is $\pm s$; Equation (22) gives b^* and κ^2. For the third moment, all vertices satisfy $x\,y\,z = s$ and every other product of three coordinates averages to zero (the four sign patterns are equally weighted), so the only non-zero entries of M are the six permutations of $\mathsf{M}_{xyz} = s$; M is already traceless ($\mathsf{M}_{cdd} = 0$), so $\big\|\mathsf{M}^0\big\|_F = \sqrt{6}\,s$, which yields J_3^*. Monotonicity of b^*: $\mathrm{d}b^*/\mathrm{d}s = 6s/(s^2+2)^2 > 0$. For J_3^*: $\mathrm{d}J_3^*/\mathrm{d}s = \sqrt{6}\,(s^2+2)^{-5/2}\,(2-2s^2)$, positive for $s<1$ and negative for $s>1$. □

Verification. The descriptors plotted in Figure 11(d) were computed numerically from the point sets Equation (25) on a grid of $s \in [1,3]$ and coincide with Equation (26) to within 3.3×10^{-16} (for example $s=2$: $b^*=\frac{1}{2}$, $J_3^*=\frac{1}{3}$, $\kappa^2=\frac{1}{4}$; $s=3$: $b^*=\frac{8}{11}$, $J_3^*=0.2014$, $\kappa^2=\frac{64}{121}$).

Why a second-moment "sphericity" is not enough.

Proposition 8.7 (Separation of shapes [proved, verified])**.** *Consider five reference configurations: the regular tetrahedron (T), the regular octahedron (O), the cube (C), a uniformly distributed sphere (Sph) and a collinear rod. Then*

(a) T, O, C and Sph all have $b^* = c^* = \kappa^2 = 0$: they are indistinguishable by R_g-based anisotropy. Only the rod has $b^* = \kappa^2 = 1$.

(b) $J_3^*(\mathrm{T}) = \sqrt{2}/3$, while $J_3^* = 0$ for O, C, the sphere and the rod: J_3^* isolates the tetrahedron (it is the only one of the five without inversion symmetry).

(c) Among the centrosymmetric isotropic shapes, Q_4 separates O ($\sqrt{7/12} = 0.764$), C ($\sqrt{7/27} = 0.509$) and Sph (0); Q_3 is non-zero only for T ($\sqrt{5}/3 = 0.745$).

Hence the vector (b^*, J_3^*, Q_4) separates all five classes.

Table 11. Shape descriptors of the reference configurations and of the real metal cores of three benchmark frameworks (metal atoms only). The numerical sphere is 2000 Fibonacci points for Q_l and 400 for the gyration quantities, hence the residual 10^{-4}. Q_3, Q_4 and Q_6 are Steinhardt invariants. "Real" rows are from the crystal structures: the Zn_4 core of MOF-5, the Zr_6 core of UiO-66 and the Cu_2 paddlewheel of HKUST-1; ZIF-8 has a single metal atom per node, and so has no shape.

Configuration	m	R_g^2	b^*	κ^2	J_3^*	Q_3	Q_4	Q_6
regular tetrahedron	4	3.000	0	0	0.471	0.745	0.509	0.629
regular octahedron	6	1.000	0	0	0	0	0.764	0.354
cube	8	3.000	0	0	0	0	0.509	0.629
sphere (numerical)	400	1.000	10^{-4}	10^{-8}	10^{-4}	10^{-5}	10^{-5}	10^{-5}
rod	41	0.350	1	1	0	0	1	1
real metal cores (distances in Å)								
Zn_4 (MOF-5)	4	3.775	10^{-16}	10^{-31}	0.471	0.745	0.509	0.629
Zr_6 (UiO-66)	6	6.139	10^{-15}	10^{-30}	10^{-17}	10^{-16}	0.764	0.354
Cu_2 (HKUST-1)	2	1.470	1	1	10^{-15}	10^{-15}	1	1

Proof. (a) Each configuration has a point group whose action on $\mathbb{R}^3$ is irreducible (T, O, C) or the full rotation group, so by Schur's lemma the symmetric second moment is a multiple of the identity. (b) A centrosymmetric point set satisfies $P = -P$, so every odd-order moment vanishes and $J_3^* = 0$; for the tetrahedron the computation of Proposition 8.6 at $s = 1$ gives $\sqrt{2}/3$. (c) Odd-l harmonics Y_{lm} change sign under inversion, so $Q_l = 0$ for odd l on centrosymmetric sets; the tetrahedron has non-zero Q_3 from its cubic invariant xyz. The values of Q_4 follow from evaluating the $l = 4$ harmonics at the six octahedron vertices and the eight cube vertices (the tetrahedron has the same Q_4 as the cube, $\sqrt{7/27}$, because its vertices are a subset of the cube's and Q_4 is even in $\hat{y}$); the sphere gives 0 by orthogonality of Y_{lm}, $l \geq 1$, to constants. □

The real building blocks. Table 11 shows that the descriptors recover the textbook shapes from crystal coordinates alone. The Zn_4 core of MOF-5 has all six Zn ⋯ Zn distances equal to 3.173 Å and is a regular tetrahedron to machine precision ($J_3^* = 0.47140$, $Q_3 = 0.7454$, i.e. $\sqrt{2}/3$ and $\sqrt{5}/3$ within 10^{-15}). The Zr_6 core of UiO-66 has twelve distances of 3.504 Å and three of $4.955 = \sqrt{2} \times 3.504$ Å, a regular octahedron ($b^* \approx 0$, $J_3^* \approx 0$, $Q_4 = \sqrt{7/12}$, $Q_6 = 1/(2\sqrt{2})$). The Cu_2 paddlewheel of HKUST-1 has a single Cu ⋯ Cu distance of 2.425 Å and is a rod ($b^* = 1$); this is degenerate, since any two-point set is collinear, and it is reported only for completeness. The instructive pair is therefore Zn_4 and Zr_6: both have an *isotropic* gyration tensor, so any second-moment criterion calls both "spherical"; they are distinguished exactly by J_3^* and by Q_3, Q_4, which is the quantitative content of the statement that they are "tetrahedral" and "octahedral".

8.3 Combining the two layers

A framework together with a choice of building-block geometry is a point $(\Gamma_S, P(s))$ of a two-parameter family: the discrete parameter S (which linkers are present) and the continuous parameter s (the shape of the node cluster). The dimension ladder of Theorem 8.3 is independent of s because ϱ depends only on the voltage graph, which ignores geometry; conversely the descriptors of Equations (22) and (23) depend only on the node cluster and not on S. This is a precise sense in which the two kinds of morphism are *orthogonal*: one can change dimensionality at fixed node shape, or node shape at fixed topology. What the model does not contain is an energy functional coupling s and S to bond lengths, angles and steric constraints: $P(s)$ for $s \neq 1$ is generally not a chemically stable cluster. Section 9 explains how the coupling would be introduced.

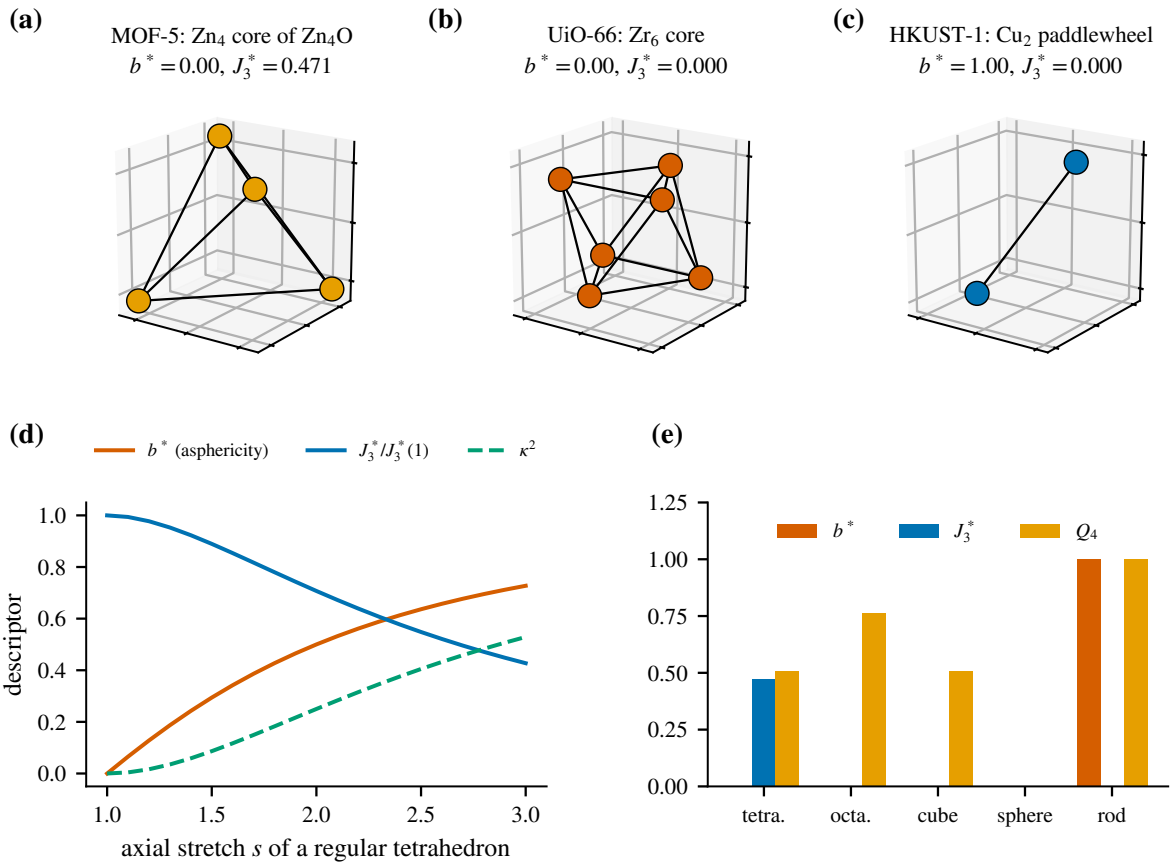


Figure 11. Shape space. (a–c) The real metal cores (spheres are metal atoms, lines join nearest metal pairs): the Zn_4 tetrahedron of MOF-5, the Zr_6 octahedron of UiO-66 and the Cu_2 paddlewheel of HKUST-1 (a two-point set, hence degenerate). (d) Descriptors along the tetrahedron-to-rod path Equation (25); legend entries are b^*, J_3^* (normalised to its value at $s = 1$) and κ^2. (e) Gyration anisotropy b^*, J_3^* and Q_4 of the five reference shapes: b^* is blind to the difference between T, O, C and the sphere, J_3^* singles out the tetrahedron and Q_4 separates the remaining isotropic shapes.

9 Discussion: what is established, what is not, and how chemistry enters

9.1 Summary of findings

Table 12 collects the claims of this paper with their status. The organising lesson is that the framework is best treated as a voltage graph (Γ, ν): the operations the project needs (building supercells, computing spectra, deleting linkers, counting components and ranks) are exact and inexpensive on that object, and what is computed on the atomic supercell can be deduced from it.

9.2 Positioning against the two reference papers

Chen et al. [4] represent a structure as an interaction complex and compute persistent interaction homology (PIH), whose persistent features become Interaction-Topological-Transformer tokens (one structural, seven elemental, up to 256 atomic and 42 pairwise tokens). The topology they use is that of a *finite* cluster of atoms. Our voltage-graph view supplies what a finite model lacks, an exact handle on periodicity: the net voltage group H, the rank ϱ and the component count of Proposition 3.5 are the homological invariants of the periodic quotient (the torus-like space $\mathbb{R}^3/\Lambda$ covered by the net), and Theorem 8.3 shows that they change in a controlled way under linker

Table 12. Status of the claims. *Proved*: a proof is given in this paper. *Verified*: recomputed on real structures. *Withdrawn*: an earlier claim that the present analysis refutes.

Claim	Content	Status
C1	Coarse-graining commutes with supercell lifting (Theorem 4.1)	proved, verified 5/5
C2	Bloch decomposition of supercell Laplacians (Theorem 5.1)	proved, verified
C3	Closed-form stiffness tensor Σ, $n^2\lambda_2 \to 4\pi^2 \min m^{\mathsf{T}}\Sigma m$ (Theorem 5.3 and Proposition 5.5)	proved, verified
C4	Multifractal spectrum shape and $D_0 = \rho_D \ln \lvert I \rvert$ (Proposition 6.2)	proved, verified
C5	Tail identity: $\Delta\alpha = \rho_D \ln N - \mu_D - \alpha(+\infty)$ (Theorem 6.3)	proved, verified
C6	Periodicity ladder $3 \to 2 \to 1 \to 0$ for MOF-5 (Theorem 8.3)	proved, verified
C7	Tetrahedron-to-rod path and separation of shapes (Propositions 8.6 and 8.7)	proved, verified
W1	Spectral width $\Delta\alpha$ as a family-specific material signature	**withdrawn**
W2	"Narrow bands" of $\Delta\alpha$ for two families	**withdrawn** (finite-size effect)

removal. A concrete *proposed* continuation is to compute persistent homology of the periodic complex on the quotient (with Betti numbers $\beta_0, \beta_1, \beta_2$ tracked along the ladder of Figure 10) and to compare it with the PIH features of [4]; we have not done this.

Anselmi et al. [1] train a multi-graph transformer (MGT) on bond, line and Coulomb graphs, screen 20 375 QMOF structures [19] down to 2 169 HSE06-level calculations and 105 hits, and report an average error of 0.34 eV across band gap, electron affinity and ionisation potential. Two points from their study bear directly on our project. First, they show that surrogate-model error can be absorbed by a *margin* δ in the screening window: on their test set, unchanged thresholds recover 20% of the DFT-selected structures, a margin equal to the model error ($\delta = 0.34$ eV) recovers 69%, and $\delta = 1.0$ eV recovers 98%, at the price of a larger follow-up pool (7%, 31% and 72% of the test set respectively). The same pre-filter logic applies to adsorption: a cheap graph-level descriptor with a known error bar can discard much of a database before an expensive simulation. Second, they find that errors depend strongly on the metal: lanthanide-containing structures have the lowest error on species seen in training (0.18 eV) but the highest on unseen ones (1.09 eV), which they attribute to memorisation of coordination geometries. A graph must therefore carry chemical identity, which our present block graph does not.

9.3 How chemical properties would be attached

The panel's central objection, that "without chemical properties nothing can be finalised", is correct, and the paper does not claim to answer it. The graph layer established here is the *structural* half of a pipeline whose second half is standard in the field; we describe it as a plan, and none of it has been run.

P1. Geometric descriptors. For each structure on a morphism path compute pore-limiting diameter, largest cavity diameter, accessible surface area and pore volume with Zeo++ (probe radius chosen for the target gas) and relate them to ϱ and s.

P2. Adsorption observables. Compute Henry coefficients $K_{H,i}$ by Widom insertion and uptake isotherms by grand-canonical Monte Carlo (RASPA) for the gases of interest with a force field (UFF/TraPPE) and EQeq or DDEC charges. Selectivity is $S_{ij} = K_{H,i}/K_{H,j}$ and working capacity is the difference of uptake between two state points.

P3. Energy along the morphism. Attach to the shape path $P(s)$ of Equation (25) an energy $E(s)$ computed with a semi-empirical or DFT method for the cluster including its ligands, making the path a minimum-energy (or constrained-optimisation) path rather than an affine map, and report the barrier.

P4. Learning. Use $(\varrho, \dim\ker \mathcal{L}_n, \Sigma,$ descriptors of Section 8.2$)$ as inputs to a surrogate for the observables of P2, with the margin-δ pre-filter of [1]; the size-independent Σ is a natural candidate input in place of $\Delta\alpha$, which Theorem 6.3 shows to be size-dependent.

9.4 Limitations

- The proofs of Theorem 4.1 and Theorem 8.3 assume finite building blocks; structures with rod-shaped inorganic building units (e.g. MIL-47-type) violate the first assumption, and the pipeline flags but does not decompose them.
- Bond perception is by covalent-radius cut-offs (Section 4) and is sensitive to borderline contacts; it inherits the limitations of MOFid [3].
- The benchmark set is four frameworks plus two collections of 72 and 55 structures. Nothing here is a statistical claim about MOFs in general.
- The shape morphism is a statement about point configurations of a node. It says nothing about the feasibility of a synthesis, and $P(s)$ for $s \neq 1$ is not in general a stable cluster.
- Interpenetrated or multi-component nets (six such cases in Section 7) need per-component treatment and have been only partially analysed.
- Chemical properties (pore geometry, adsorption, selectivity) have not been computed.

9.5 Outlook

The most valuable next step is P1–P2 on the four benchmark frameworks and the eight states of the MOF-5 ladder (the 3D framework, the three 2D and the three 1D sub-nets obtained by choosing which axes to keep, and the 0D state; Figure 10 shows one representative of each rank), which would give the first data set linking the mathematics of Section 8 to a measurable property. A second is the study of how Σ, which is cell-independent and computable in $\mathcal{O}(|V|^3)$ from the voltage graph, correlates with guest diffusivities computed by molecular dynamics. A third is to extend Theorem 8.3 from the pcu-type example to arbitrary nets by characterising, for any (Γ, ν), the lattice of achievable rank vectors under linker deletion, a problem closely related to cut spaces of graphs with group-valued labels [12].

Data and code availability

All code, data and figures are in the accompanying archive (`code/` and `paper/`). Appendix A maps every numerical claim in this paper to the script that produces it. The four test structures (HKUST-1, MOF-5, ZIF-8, UiO-66) are included in the archive; the hMOF and CoRE MOF records were fetched from MOFX-DB [2].

Acknowledgements

We thank the project panel, whose objection that connectivity-only results are not final without chemical properties, and whose request for a mathematical and computational treatment of framework morphism, shaped Sections 8 and 9.

A Reproducibility map and algorithms

A.1 Notation

Symbol	Meaning
(Γ, ν), G_n	voltage graph; n-supercell (Definition 3.1)
$\lvert V\rvert, \lvert E\rvert, \beta_1$	base vertices, edges, cyclomatic number
H, ϱ	net voltage group, periodicity rank
$\mathcal{L}(G)$, $\mathcal{L}(k)$	graph Laplacian; Bloch matrix Equation (7)
Σ	stiffness tensor Equation (10)
N, D	number of vertices of G_n; diameter
$p_i(r)$, $Z_q(r)$	box measure; partition function Equation (13)
$\tau, \alpha, f, \Delta\alpha$	mass exponent, singularity exponent, spectrum, width
ρ_D, μ_D	Equation (16), Equation (17)
$b^*, \kappa^2, J_3^*, Q_l$	shape descriptors Equations (22) and (23)

A.2 Algorithms

Algorithm 2 Periodicity rank of a component (used for Theorem 8.3).

Require: component C of a voltage graph, spanning tree T found by depth-first search from a root

1: set $\pi(\text{root}) = 0$ and $\pi(v) = \pi(u) + \nu(u \to v)$ along tree edges
2: for every non-tree edge $e = (u \to v)$ set $w_e = \nu(e) + \pi(u) - \pi(v) \in \mathbb{Z}^3$ ▷ fundamental-cycle voltage
3: stack the w_e into a matrix W and return $\varrho = \operatorname{rank} W$

The vectors w_e generate H because the fundamental cycles generate the cycle space and the voltage is additive along closed walks. $\beta_1 = \lvert E\rvert_C - \lvert V\rvert_C + 1$ is the number of rows of W.

Algorithm 3 Stiffness tensor and supercell gap (Theorem 5.3).

Require: labelled quotient graph (Γ, ν) with $\lvert V\rvert$ vertices and $\lvert E\rvert$ edges

1: assemble $B \in \mathbb{R}^{V \times E}$ and $T \in \mathbb{R}^{E \times 3}$; $\mathcal{L} = BB^{\mathsf{T}}$; $g = BT$
2: $\Sigma_{\text{lat}} = \left(T^{\mathsf{T}}T - g^{\mathsf{T}}\mathcal{L}^{+}g\right) / \lvert V\rvert$ ▷ $\mathcal{O}(\lvert V\rvert^3)$ via an eigendecomposition
3: $\Sigma_{\text{cart}} = M^{\mathsf{T}}\Sigma_{\text{lat}}M$
4: predicted gap: $n^2\lambda_2 \to 4\pi^2 \min_{m \in \mathbb{Z}^3 \setminus 0} m^{\mathsf{T}}\Sigma_{\text{lat}}m$ (minimum over a small integer box)

A.3 Map from results to scripts

All scripts are in the `code/` directory of the accompanying archive, are deterministic (fixed seeds) and write one JSON file each; `make_paper_figures.py` reads those files and produces every figure.

Random seeds are fixed in every script; the multifractal results use 20 coverings unless otherwise stated. The stored data sets in `code/src/.../9_dataset/size_scaling/` (supercell sweeps and the two large-sample runs, Section 7.3) are inputs, not outputs, of these scripts. The CIF files for the four benchmark frameworks are included.

Algorithm 4 Multifractal spectrum by box covering (Definition 6.1).

Require: supercell graph G, influential set I, number of trials S, grid $q \in [-10, 10]$

1: **for** $s = 1, \ldots, S$ and $r = 1, \ldots, D$ **do**
2: greedy random box covering of the *full* graph with boxes of radius r; $p_i^{(s)}(r) = |B(i)| / N$ for $i \in I$
3: **end for**
4: **for** each q **do**
5: $y_k(q) = \frac{1}{S} \sum_s \ln \sum_{i \in I} p_i^{(s)}(r_k)^q$ ▷ log *before* averaging over trials
6: $\tau(q) = \sum_k x_k y_k / \sum_k x_k^2$, $x_k = \ln(r_k / D)$
7: **end for**
8: $\alpha = \tau'$ (central differences), $f = q\alpha - \tau$

Table 13. Where each number in the paper comes from.

Script	Produces	Used in
`check_01_graph_defects.py`	degrees, λ_2 of cell and supercell, defect effects, centrality classes	Tables 3 and 5
`check_02_bloch_tail.py`	Bloch vs explicit spectra, $\lambda_2(\mathcal{L}_n)$ for $n \leq 16$, stored tail slopes	Theorem 5.1, Figure 3
`check_03_morphism_ladder.py`	ranks, components, β_1 along the MOF-5 ladder; zero multiplicity; 200-deletion interlacing test	Theorem 8.3 and Proposition 3.6
`check_04_shapes_band.py`	reference-shape descriptors, tetrahedron stretch path, real node cores, full-band correlations	Tables 8 and 11
`check_05_tau_spectra.py`	Steinhardt Q_l, τ demonstration, spectra of four nets	Figures 5 and 7, Table 7
`check_06_stiffness.py`	Σ tensors, Betti numbers, ρ_D, cell/supercell invariance	Table 4
`check_07_tail_identity.py`	$\alpha(-10)$, $\alpha(-\infty)$, $\rho_D \ln N$, μ_D, widths	Table 6, Figure 6
`check_08_ranks_cores_fits.py`	rank of all four nets, finite-size fits from the stored sweep, real-core distances and Q_l	Table 3, Table 11, Figure 6
`make_paper_figures.py`	all figures	Figures 3 to 11

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